

Renesas RA Family

CPK-RA8T2 EtherCAT CiA402 SubDevice Software

介绍

本文档介绍了如何使用RA8T2 移植EtherCAT 协议栈代码实现从站IO和CiA402功能，同时介绍如何使用TwinCAT主站进行测试。

1. 概述	3
1.1 缩写/定义	3
2. 软件硬件要求	4
3. 硬件	5
3.1 CPK-RA8T2 Board.....	5
4. 主站设置	6
4.1 拷贝ESI文件.....	6
4.2 安装驱动	6
4.3 生成EtherCAT从站源码	7
4.4 e2 studio加载工程	10
5. 连接主站	15
5.1 连接TwinCAT3.....	15
5.2 扫描网络.....	15
5.2.1 更新EEPROM.....	17
5.3 IO功能验证.....	18
5.4 CiA402 测试.....	20
5.4.1 OP状态确认.....	20
5.4.2 CiA402 状态切换	20
5.4.3 csp Mode	23
5.4.4 csv Mode.....	24

6. CiA402 协议	26
6.1 操作模式	26
6.2 状态切换	27
6.3 对象字典	27
6.4 CiA402相关API	29
7. 附录	35
7.1 附录 A:使用SSC工具注意事项	35
7.2 附录 B: 怎么安装patch工具	37
7.3 附录 C:重新创建工程的FSP配置步骤（不导入本样例程序）	38

1. 概述

本文档介绍了如何使用RA8T2 移植EtherCAT 协议栈代码实现从站IO和CiA402功能

1.1 缩写/定义

1	CoE	CAN application protocol over EtherCAT
2	CiA	CAN in Automation
3	DC	Distributed Clock
4	EEPROM	Electrically Erasable Programmable Read-Only Memory
5	EoE	Ethernet over EtherCAT
6	ESC	EtherCAT SubDevice Controller
7	ESI	EtherCAT SubDevice Information
8	ESM	EtherCAT State Machine
9	ETG	EtherCAT Technology Group
10	FoE	File Access over EtherCAT
11	PDO	Process Data Object
12	SDO	Service Data Object
13	SSC	SubDevice Stack Code
14	FSP	Flexible Software Package
15	IDE	Integrated Development Environment
16	GCC	GNU Compiler Collection

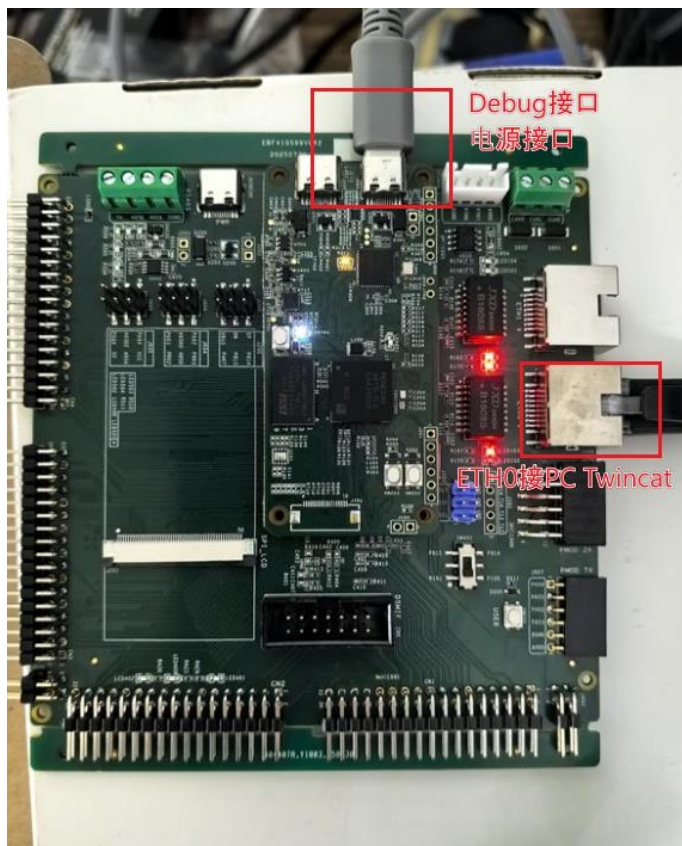
2. 软件硬件要求

Category	Name	Version	Description
Board	CPK-RAT2	-	Renesas CPK-RAT2
IDE	e ² studio	2026-04.2	Renesas RA Flexible Software Package (FSP)
Configurator	FSP Smart Configurator	2026-04.2	
Flexible Software Package	FSP for Renesas	v6.5.0	
LLVM Compiler	LLVM Embedded Toolchain for Arm	21.1.1	
Emulator	J-Link™	8.60	SEGGER SEGGER - The Embedded Experts - Downloads - J-Link / J-Trace
Software	SSC Tool	5.13	Beckhoff Automation ET9300 EtherCAT Slave Stack Code Beckhoff Worldwide
	TwinCAT3	4026.19	Beckhoff Automation TwinCAT 3.1 Build 4026 Beckhoff Worldwide
	Tera Term	5.5.1	Tera Term Tera Term Open Source Project

3. 硬件

本节介绍了使用的主要硬件。有关开发板的更多信息，请参阅各自的评估板用户手册和原理图。

3.1 CPK-RA8T2 Board



4. 主站设置

4.1 拷贝ESI文件

在启动TwinCAT3 之前，把文件目录里的 ESI 文件（xml）复制到 TwinCAT 目标文件夹里。

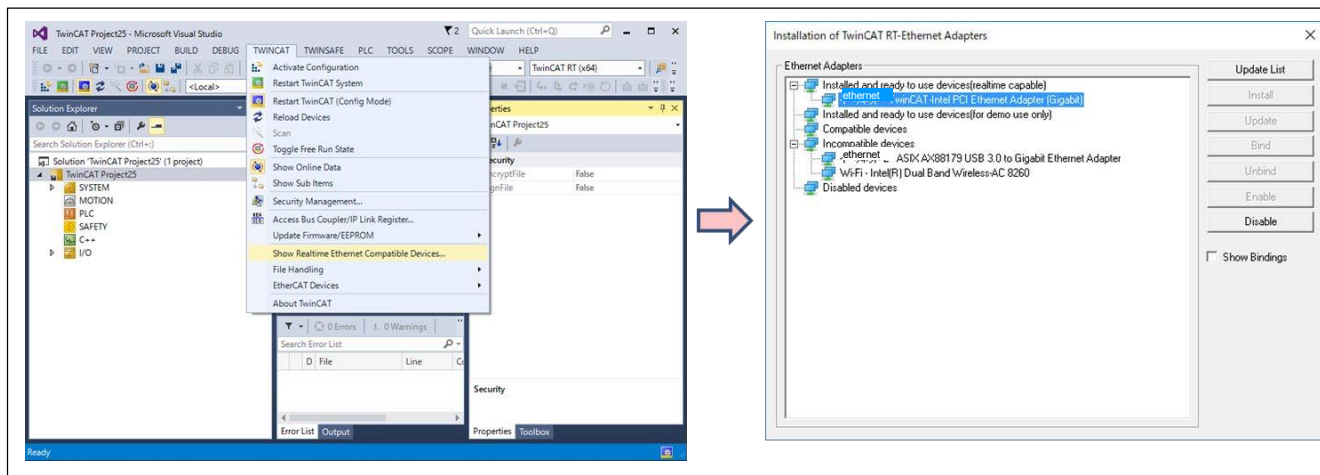
ESI文件所在目录：cpk-ra8t2-ethercat-package\common\CiA402\ESI

ESI文件名：Renesas EtherCAT RA8 IO CiA402.xml

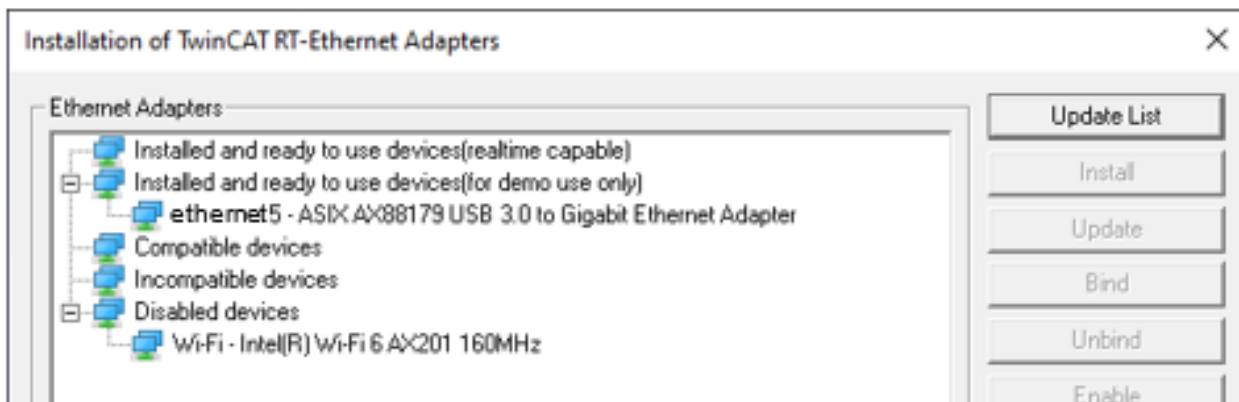
将ESI拷贝到TwinCAT的安装目录：TwinCAT\3.x\Config\IO\EtherCAT

4.2 安装驱动

TwinCAT中添加EtherCAT 驱动(首次安装).



注意：如果电脑使用的非intel网卡，可能会显示安装在（for demo use only）



4.3 生成EtherCAT从站源码

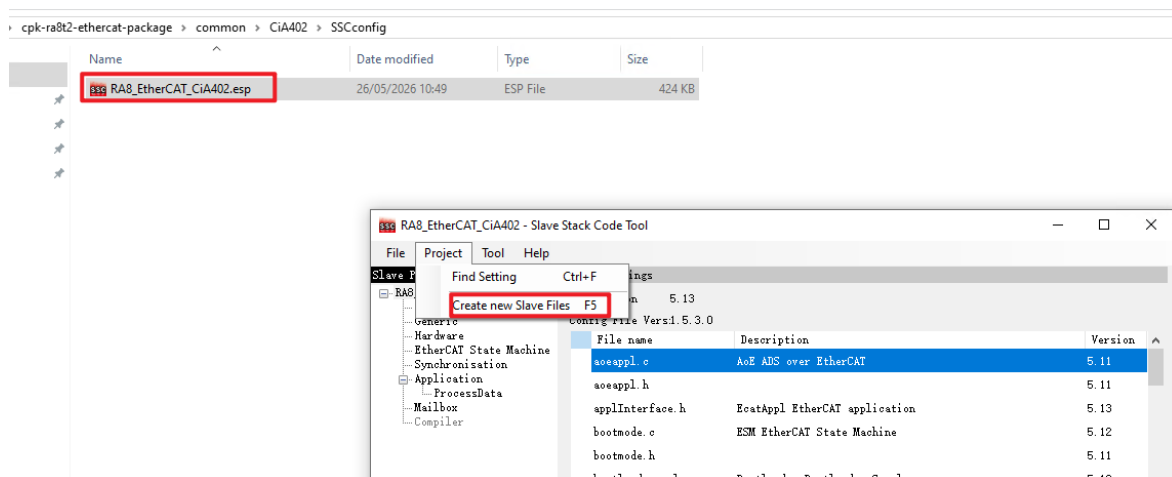
本章介绍如何使用 SSC 工具生成 EtherCAT 从站协议栈代码。

SSC工具需要从ETG官网下载

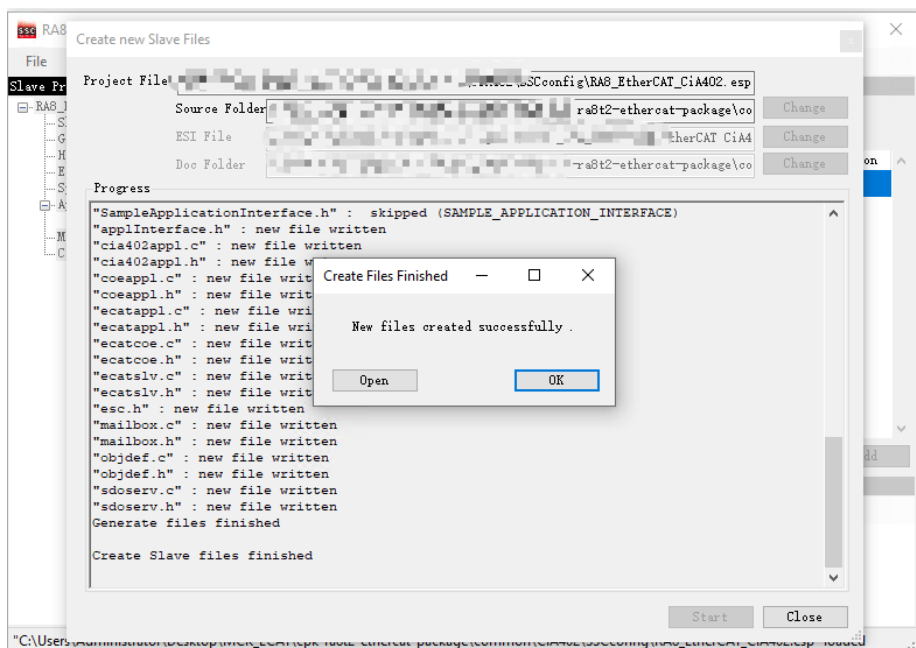
[EtherCAT Technology Group | EtherCAT Slave Stack Code \(SSC\) ET9300](#)

1. 双击该路径下的esp工程，点击> Create new Slave Files

esp工程文件所在路径：cpk-ra8t2-ethercat-package\common\CiA402\SSCconfig



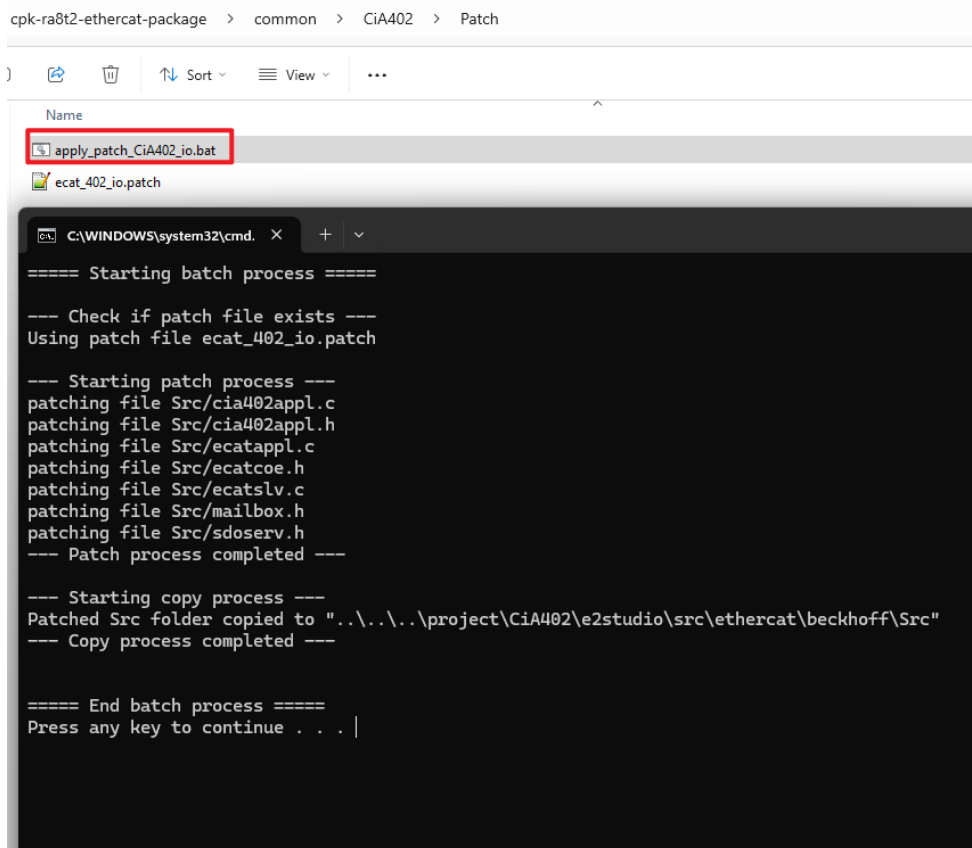
2. 点击[Start]生成EtherCAT源码，当出现New file created successfully，表示cpk-ra8t2-ethercat-package\common\CiA402\SSCconfig路径下已生成源码



» cpk-ra8t2-ethercat-package » common » CiA402 » SSCconfig » Src

Name	Date modified	Type
applInterface.h	10/08/2026 21:41	H File
cia402appl.c	10/08/2026 21:41	C File
cia402appl.h	10/08/2026 21:41	H File
coeappl.c	10/08/2026 21:41	C File
coeappl.h	10/08/2026 21:41	H File
ecat_def.h	10/08/2026 21:41	H File
ecatappl.c	10/08/2026 21:41	C File
ecatappl.h	10/08/2026 21:41	H File
ecatcoe.c	10/08/2026 21:41	C File
ecatcoe.h	10/08/2026 21:41	H File
ecatslv.c	10/08/2026 21:41	C File
ecatslv.h	10/08/2026 21:41	H File
esc.h	10/08/2026 21:41	H File
mailbox.c	10/08/2026 21:41	C File
mailbox.h	10/08/2026 21:41	H File
objdef.c	10/08/2026 21:41	C File
objdef.h	10/08/2026 21:41	H File
sdoserv.c	10/08/2026 21:41	C File
sdoserv.h	10/08/2026 21:41	H File

3. 双击批处理文件对EtherCAT从站代码打补丁，显示如下表示打补丁成功
cpk-ra8t2-ethercat-package \common\CiA402\Patch\ apply_patch_CiA402.bat



4. 此时EtherCAT源码被拷贝到工程目录。

Source folder: (code generated in the Src folder by SSC)

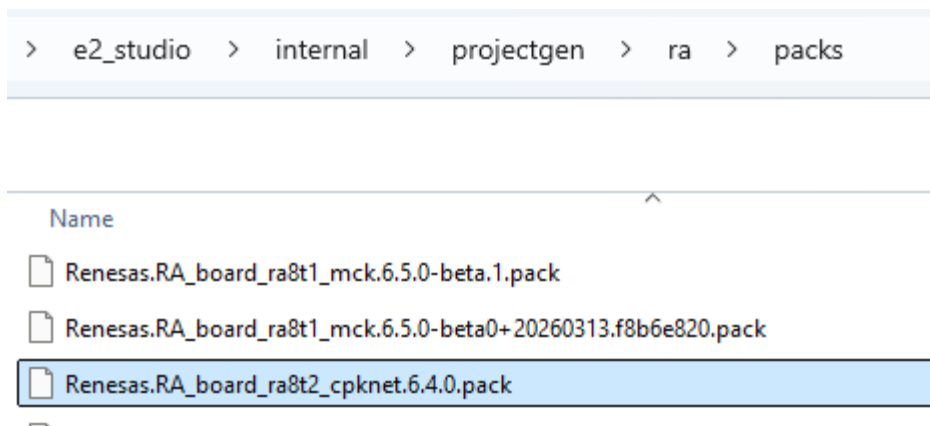
"cpk-ra8t2-ethercat-package \common\CiA402\SSCconfig\Src"

Copy destination folder:

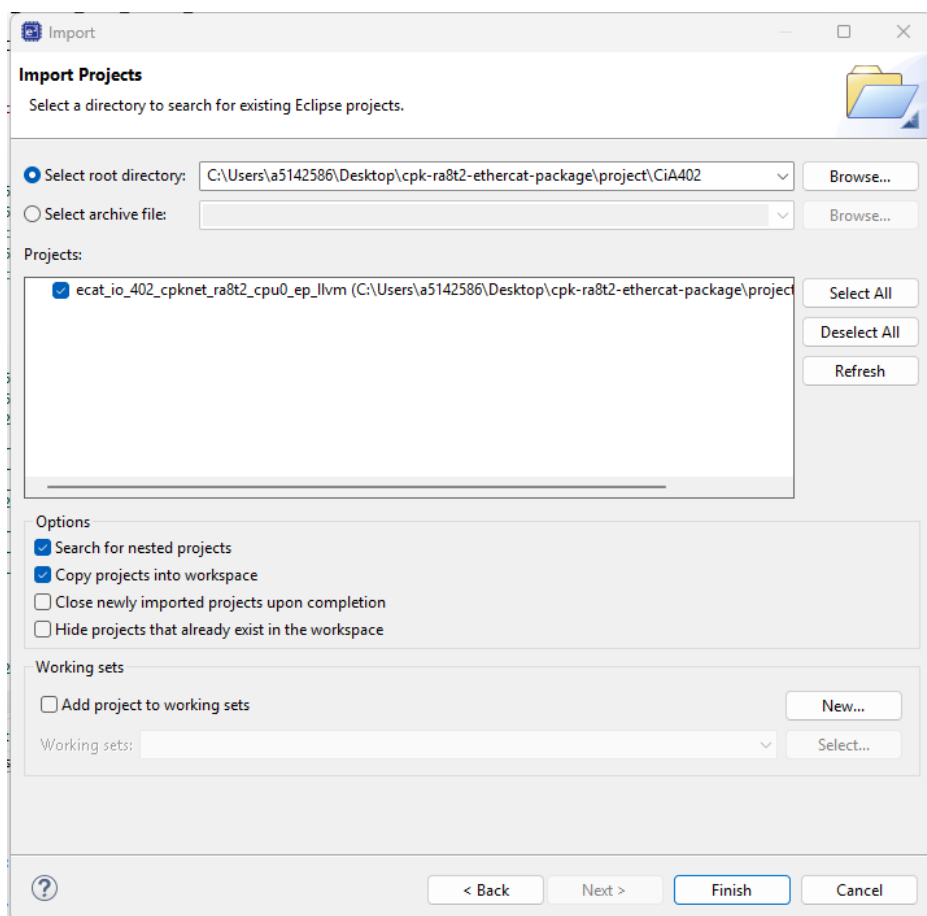
"cpk-ra8t2-ethercat-package \project\CiA402\e2studio\src\ethercat\beckhoff\Src"

4.4 e2 studio加载工程

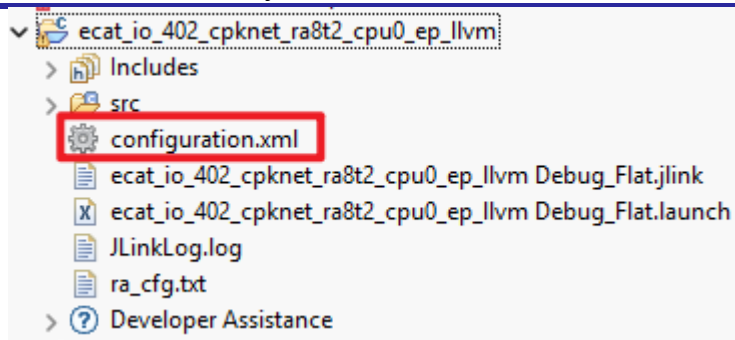
1. 安装CPK RA8T2 BSP文件，将Renesas.RA_board_ra8t2_cpknnet.6.4.0.pack文件拷贝到e2 studio的安装目录中。



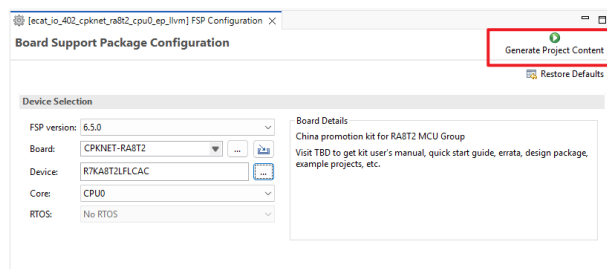
2. 打开e2 studio，导入工程。选择File→Import→Existing Projects into Workspace。选择“Select root directory”，选择cpk-ra8t2-ethercat-package \project\CiA402点击[Finish].



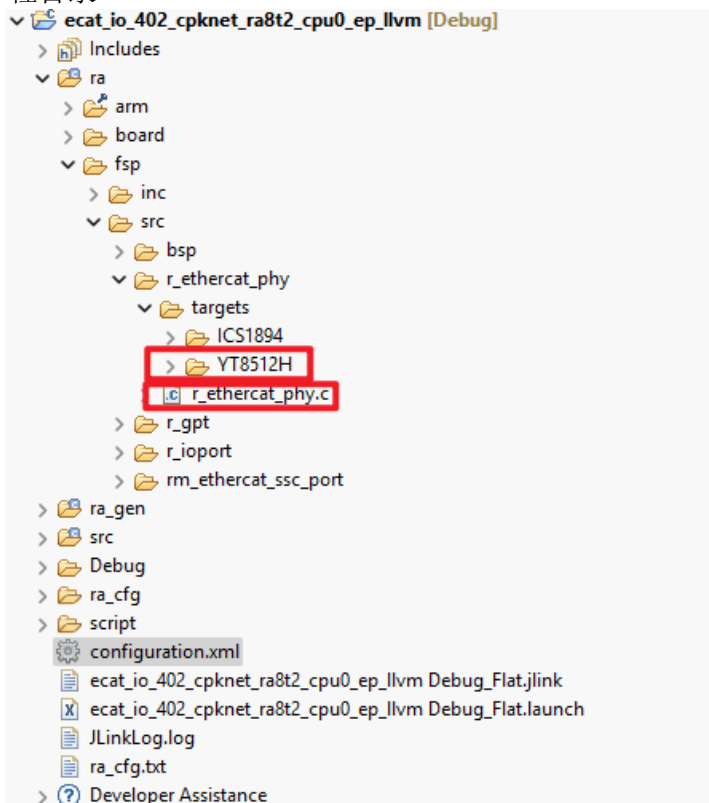
3. 打开 "configuration.xml"



4. 点击 “Generate Project Content”生成FSP配置代码



5. 将cpk-ra8t2-ethercat-package\common\ReplacementFiles\ra目录中的PHY配置文件拷贝覆盖到工程目录

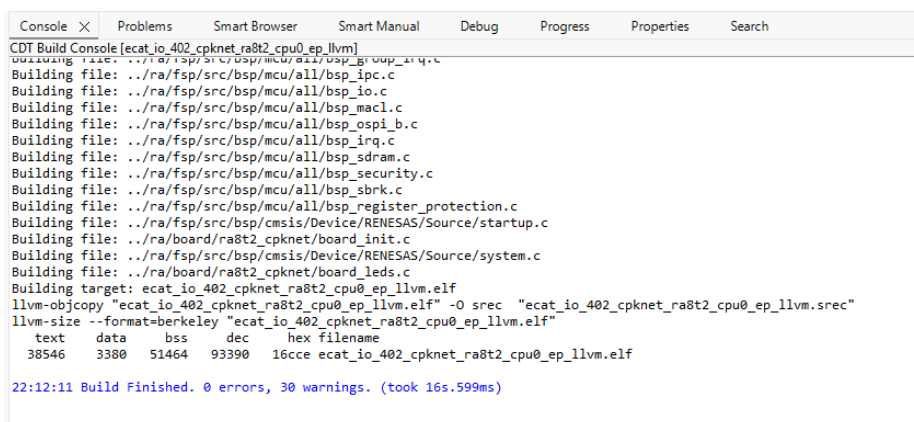
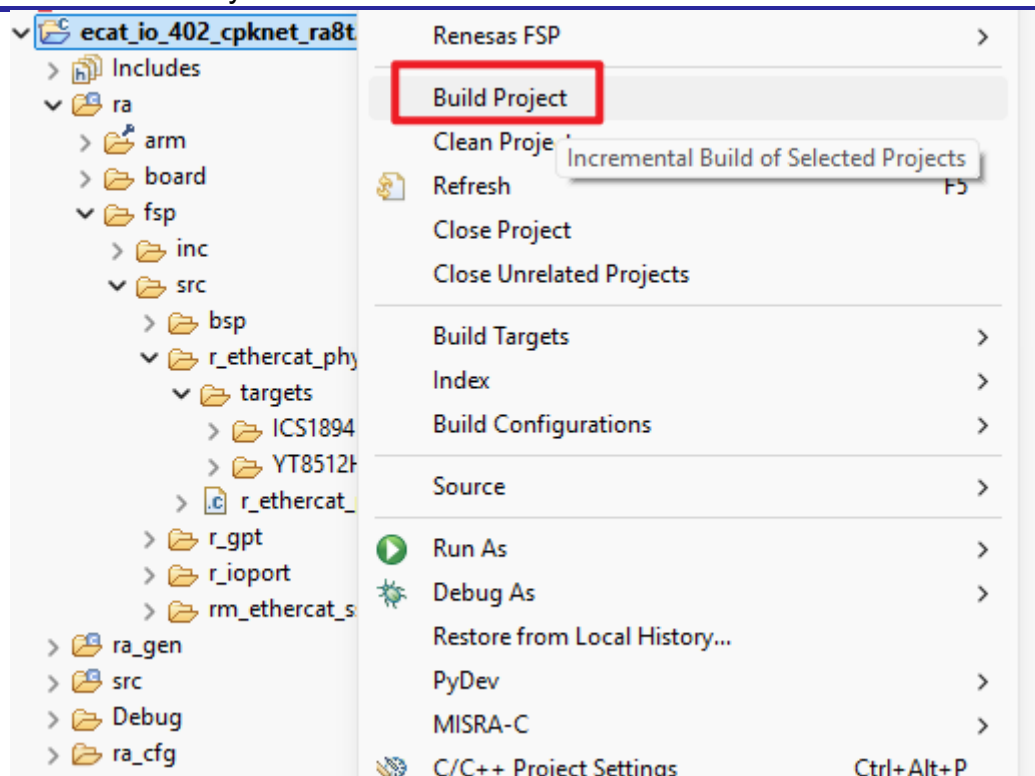


手动修改r_ethercat_phy.c文件如下（如果后续更新FSP版本生成的代码已修改，则无需手动修改）

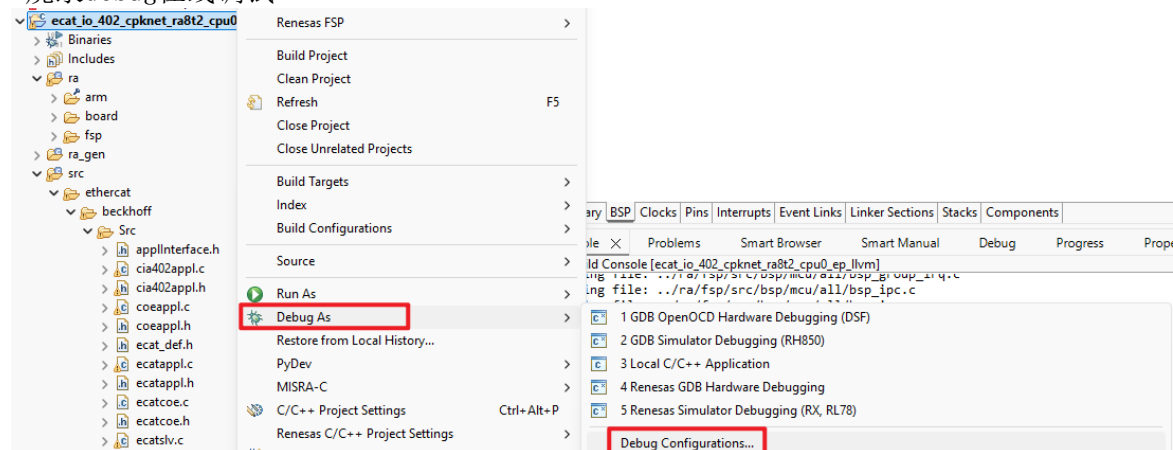
```
#if (ETHER_PHY_CFG_USE_CUSTOM_PHY_LSI_ENABLE)
case ETHER_PHY_LSI_TYPE_CUSTOM:
{
    if (NULL != p_instance_ctrl->p_ether_phy_cfg->p_extend)
    {
        ethercat_phy_extended_cfg_t const * p_callback = p_instance_ctrl->p_ether_phy_cfg->p_extend;
        if (NULL != p_callback->p_target_init)
        {
            p_callback->p_target_init(p_instance_ctrl);
        }
    }
}

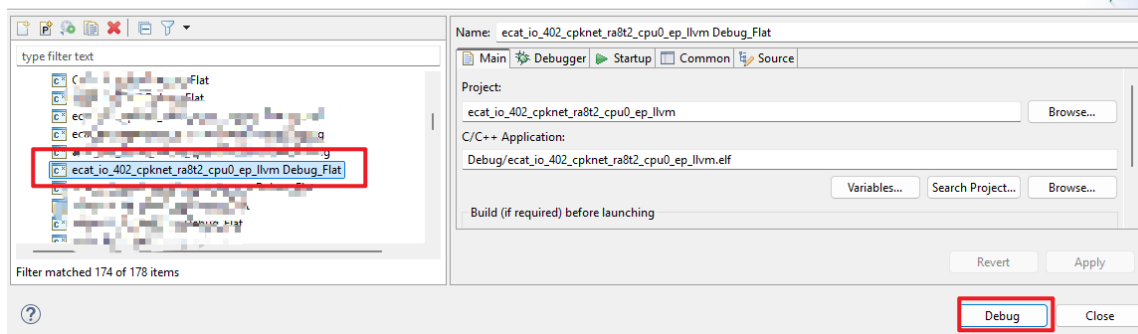
#if (ETHER_PHY_CFG_USE_CUSTOM_PHY_LSI_ENABLE)
case ETHER_PHY_LSI_TYPE_CUSTOM:
{
    if (NULL != p_instance_ctrl->p_ether_phy_cfg->p_extend)
    {
        ethercat_phy_extended_cfg_t const * p_callback = p_instance_ctrl->p_ether_phy_cfg->p_extend;
        if (NULL != p_callback->p_target_link_partner_ability_get)
        {
            result = p_callback->p_target_link_partner_ability_get(p_instance_ctrl, line_speed_duplex);
        }
    }
    break;
}
#endif
```

6. 编译工程



7. 烧录debug在线调试





5. 连接主站

CPK板子重新上电，ETH0通过网线连接TwinCAT主站

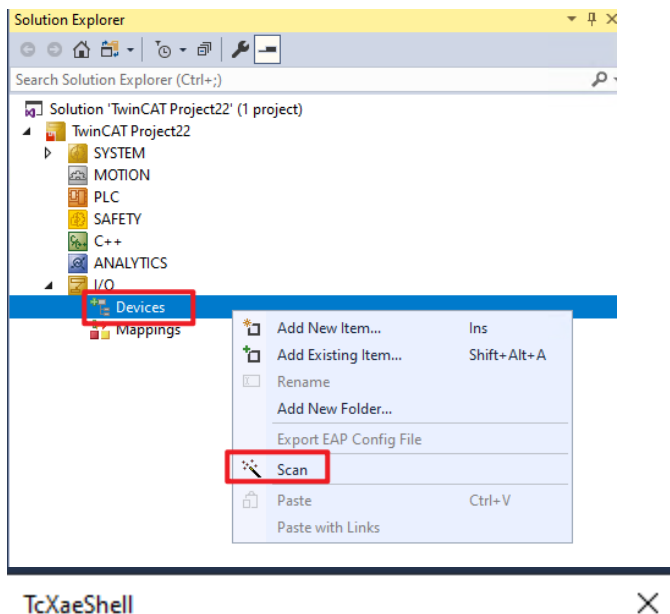
5.1 连接TwinCAT3

开始菜单打开TwinCAT软件，点击Beckhoff→TwinCAT3→TwinCAT XAE。

软件打开后，选择File → New → Project创建新工程。

5.2 扫描网络

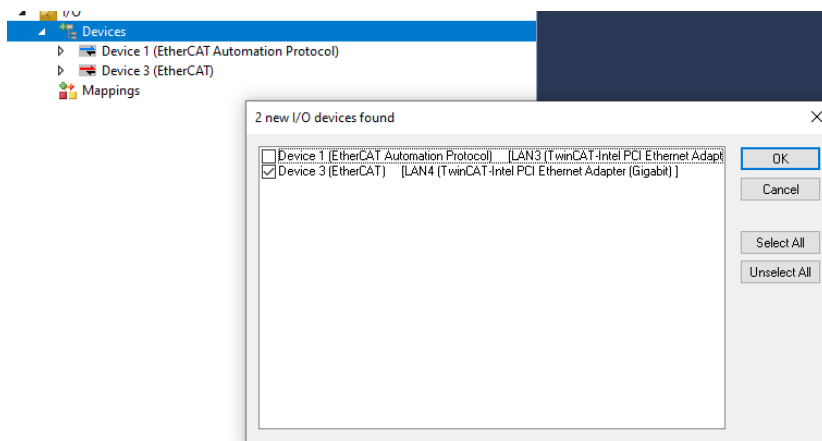
1. [I/O] → [Devices], select “Scan” as in Figure below.



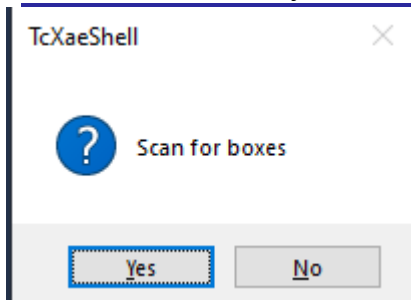
HINT: Not all types of devices can be found automatically



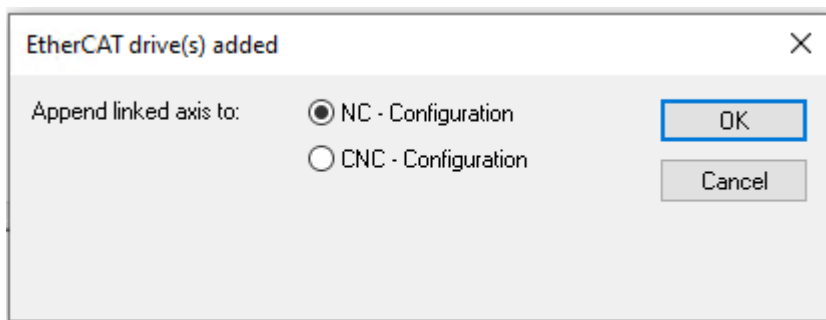
2. 选择网卡，点击ok。



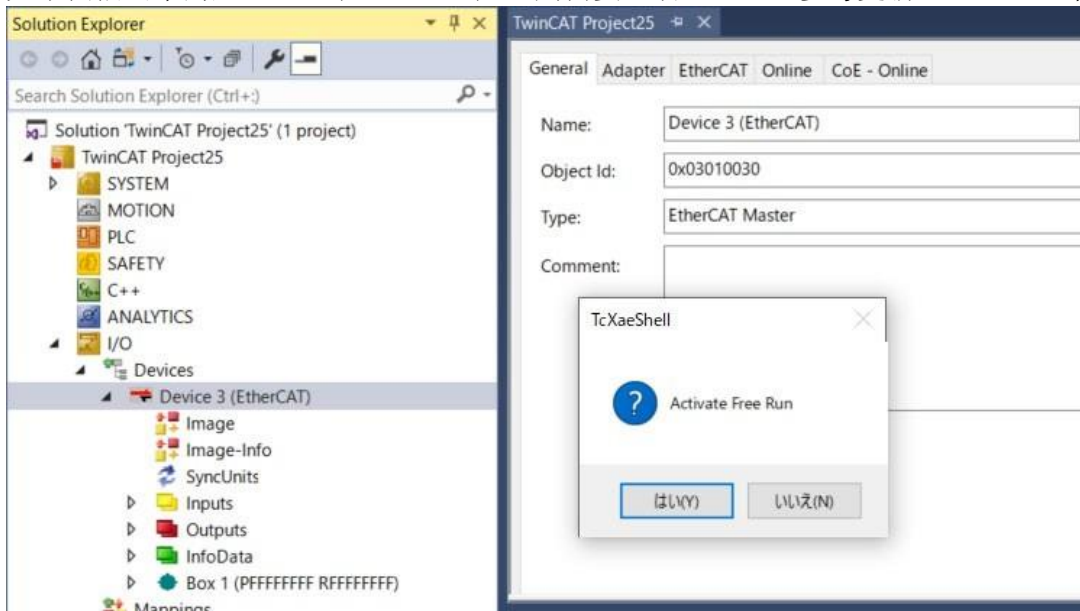
选择yes。



选择OK

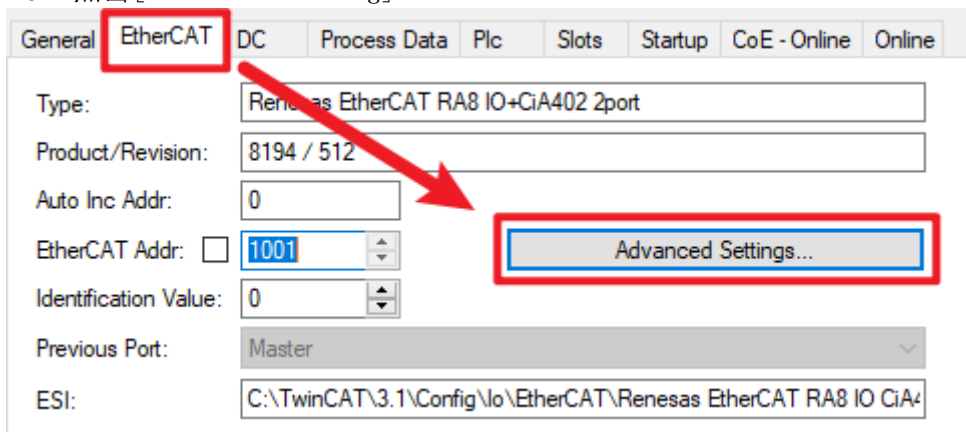


如果扫描出来的是“Box1 (PFFFFFFF RFFFFFFF)”，则需要重刷EEPROM，参考更新EEPROM这一章节。

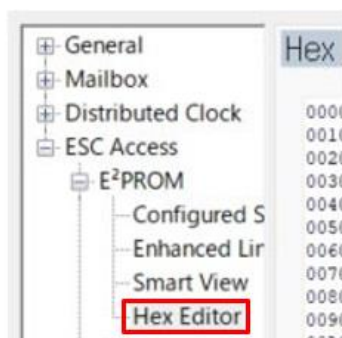


5.2.1 更新EEPROM

1. 双击[Box 1]展开菜单栏
2. 选择[EtherCAT]
3. 点击[Advanced Setting]



4. 选择[ESC Access]→[EEPROM]→[Hex Editor]



5. 点击[Download from List]



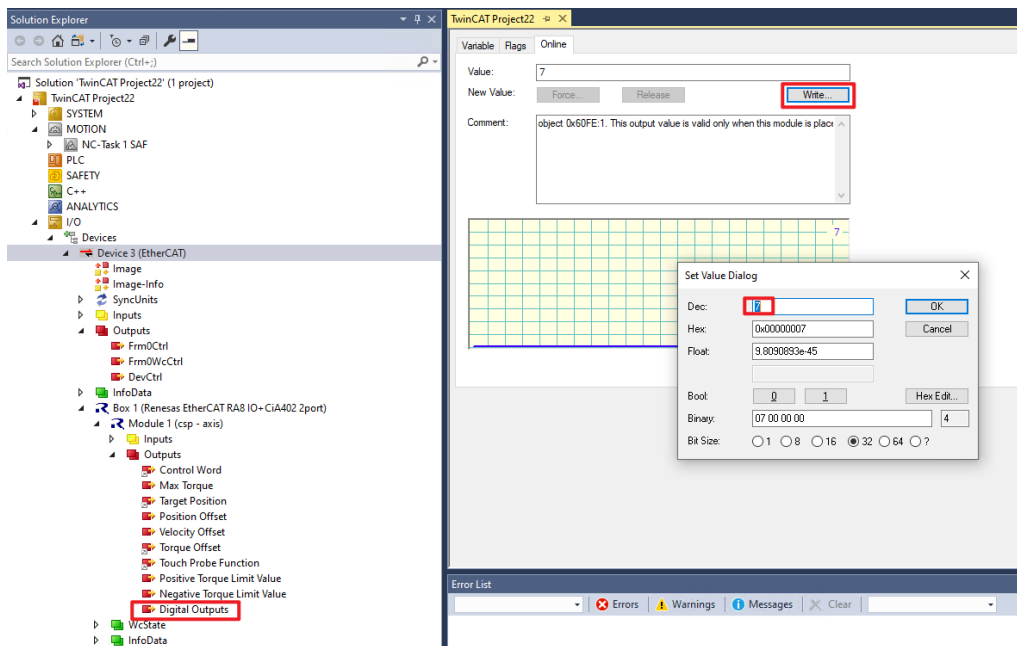
6. 选择ESI文件，点击OK开始刷EEPROM，之后重新扫描从站。

Write EEPROM

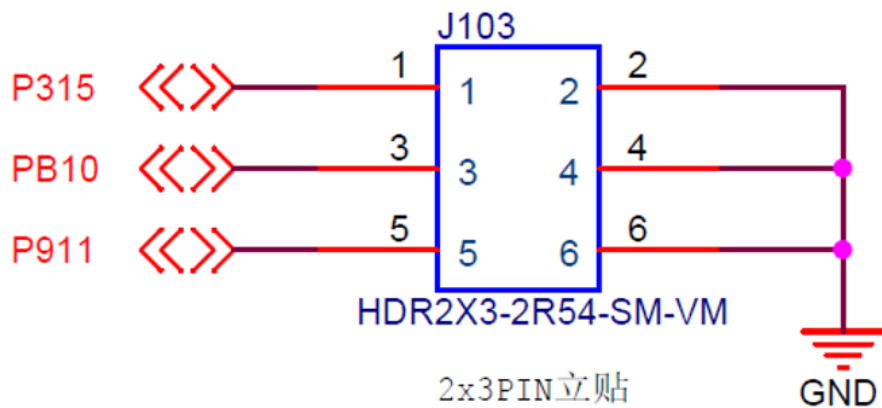


5.3 IO功能验证

IO输出功能可以通过LED400/LED401/LED402，写7表示全亮



IO输入功能可以通过J103跳线帽接地

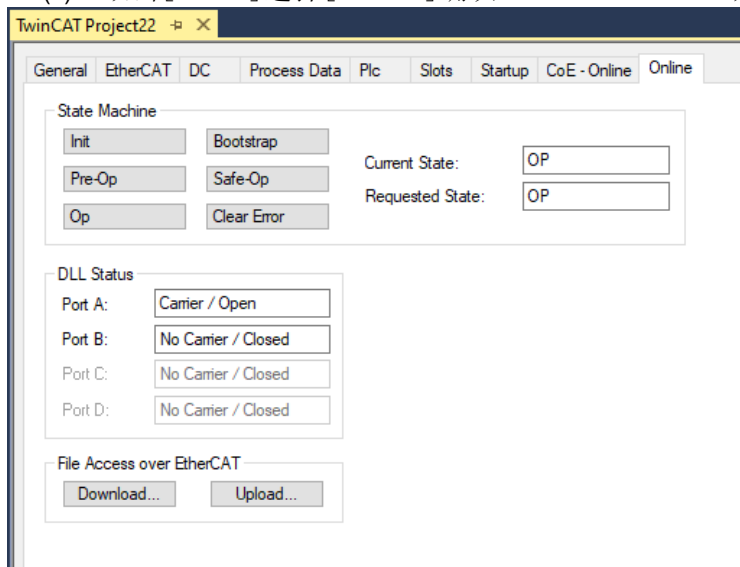


The screenshot displays the TwinCAT Project22 software interface. On the left, the **Solution Explorer** shows a project tree for 'TwinCAT Project22'. The tree is expanded to show the **Devices** folder, which contains **Device 3 (EtherCAT)**. Under this device, the **Inputs** folder is expanded, showing a list of input variables including **Digital Inputs**, which is highlighted with a red box. On the right, the **Variable** declaration window is open, showing the declaration of a variable named **14** (highlighted with a red box). The **Value** field is set to **object 0x60FD:0**. Below the variable declaration, there is a grid area. At the bottom of the interface, the **Error List** is visible.

5.4 CiA402 测试

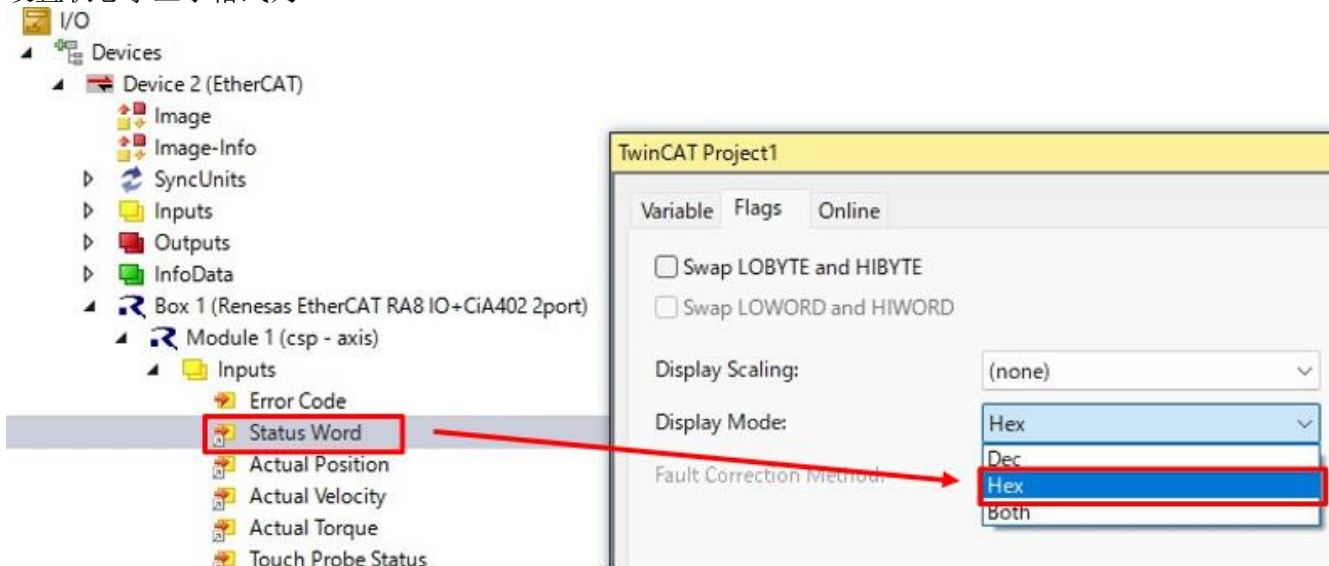
5.4.1 OP状态确认

(1) 双击[Box 1]选择[Online]确认“Current Status”是“OP”。

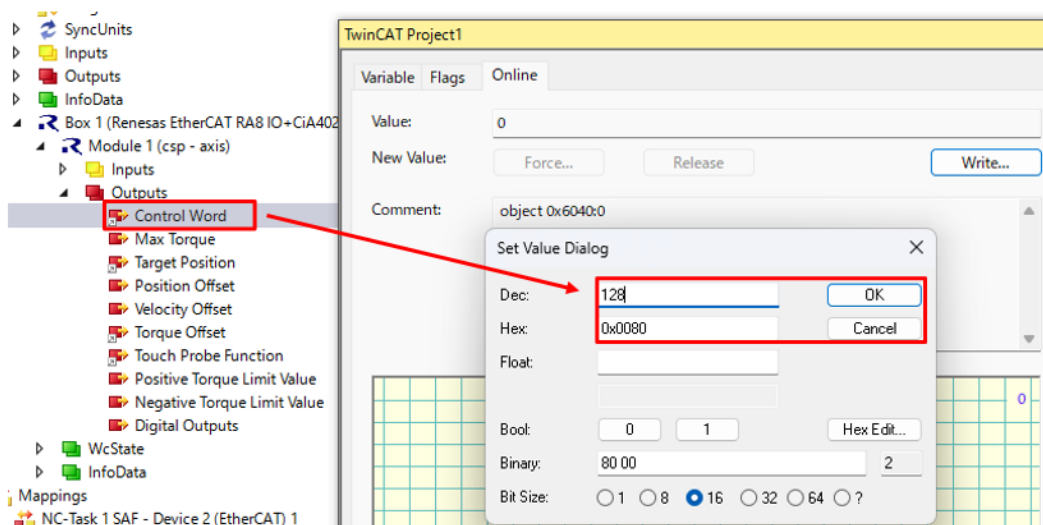


5.4.2 CiA402 状态切换

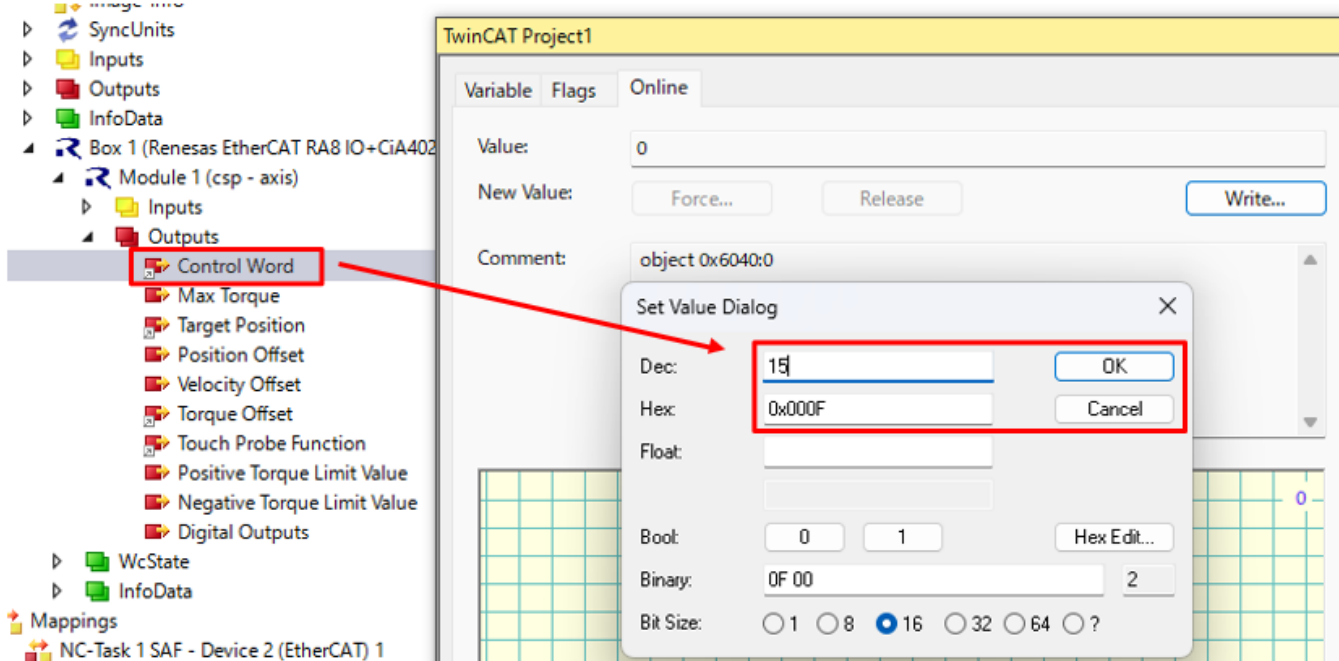
(1) 设置状态字显示格式为HEX



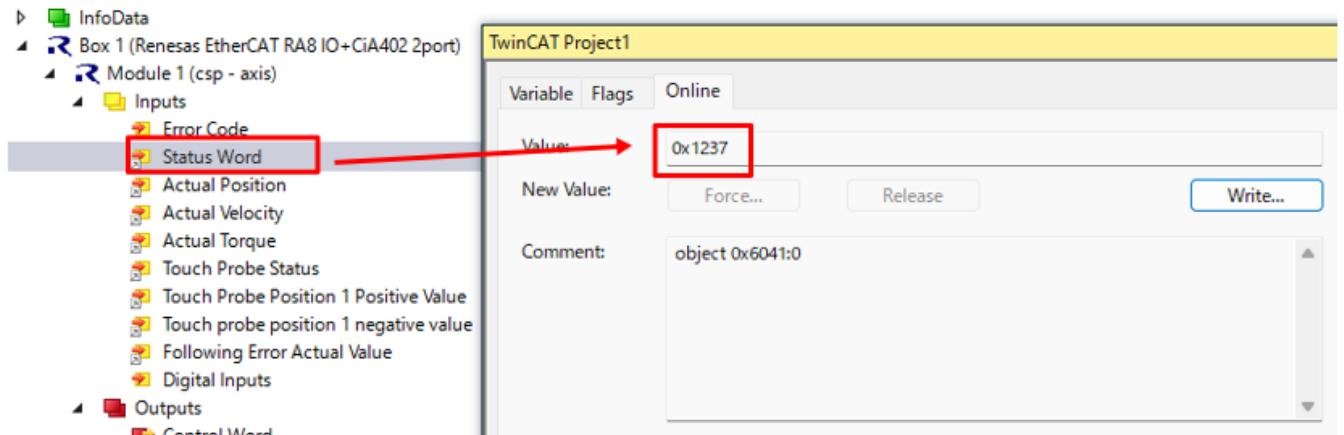
(2) 控制字写 128 (Dec)。



(3) 控制字依次写7，写15



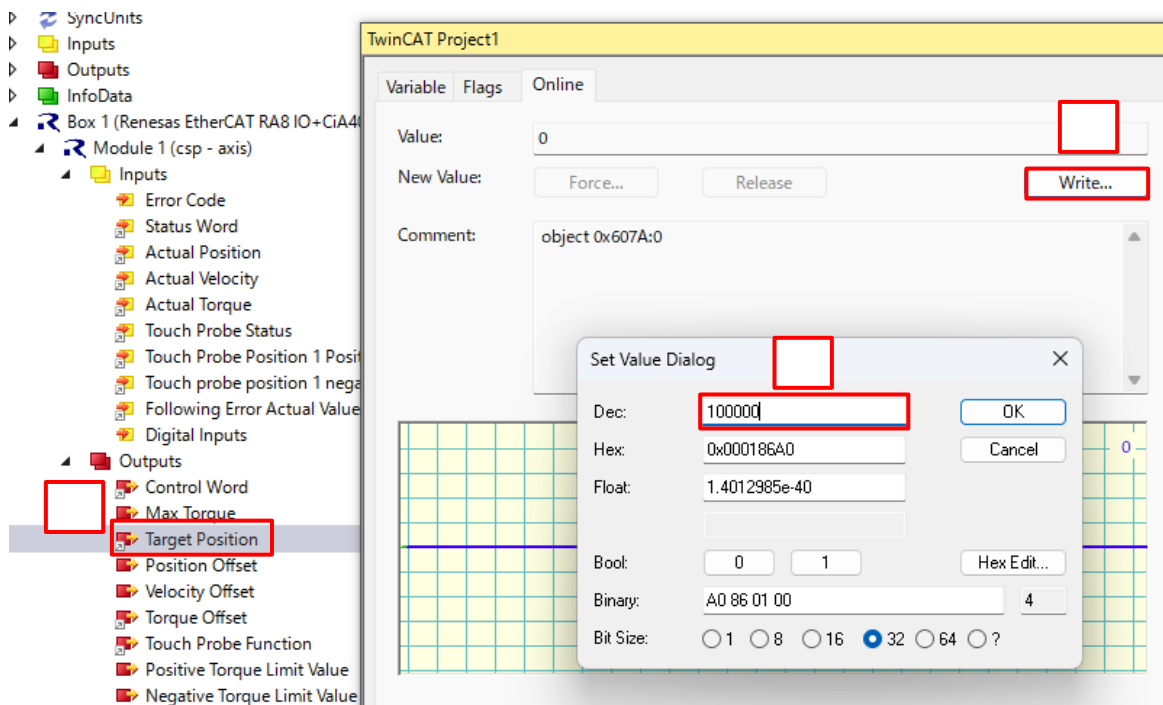
- (4) 此时状态字为0x1237表示进入“Operation Enabled”。可进一步操作
- (5) 如果此时状态字为[0x1208], 表示状态为“Fault”, 需要重新对控制字写128->7->15。



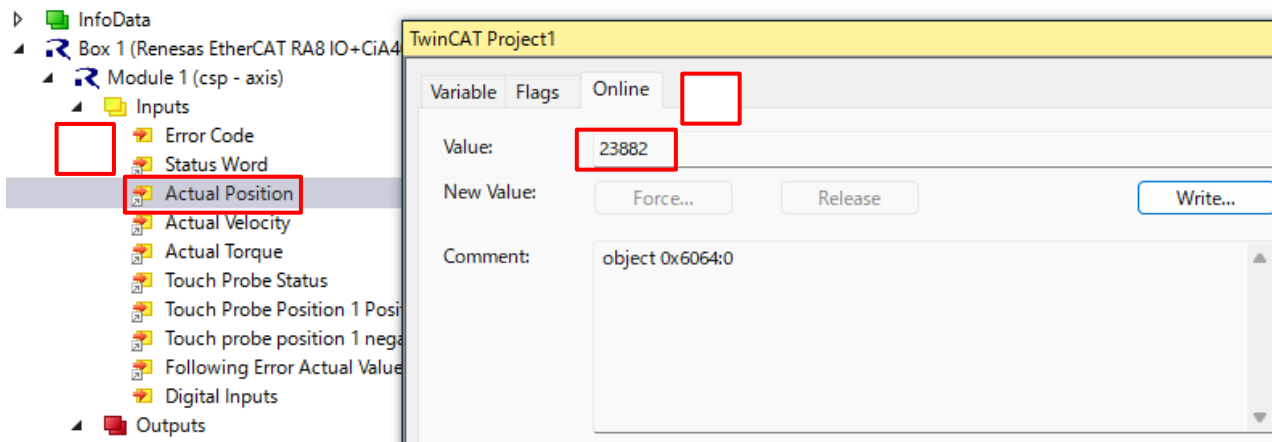
5.4.3 csp Mode

此时“Module 1”处于csp模式。

(1) 写Target Position值为100000



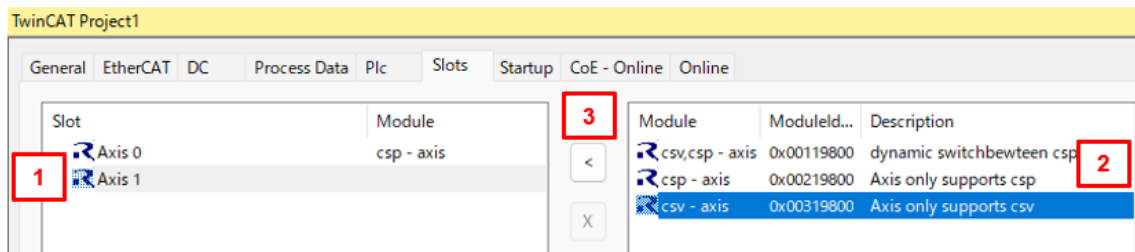
(2) 观察 “Actual Position”变化到100000



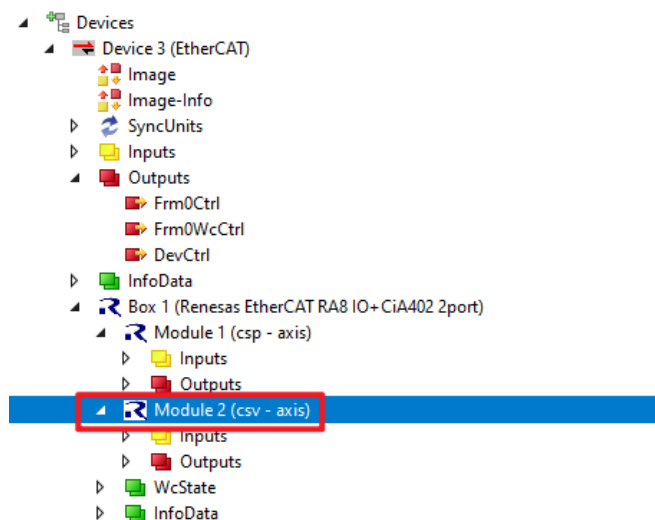
5.4.4 csv Mode

修改操作模式。

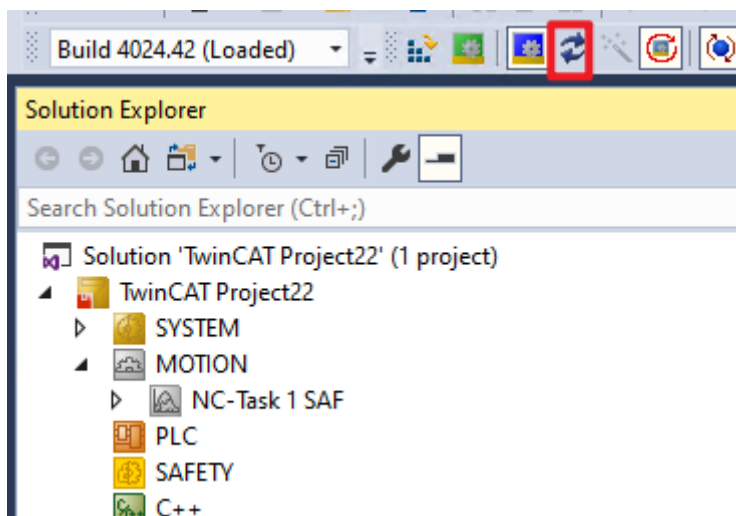
- (1) 点击 “Box 1(Renesas EtherCAT RA8 IO+CiA402 2port)”选择[Slots]
- (2) 选择 “Axis 1”添加“csv-axis”。



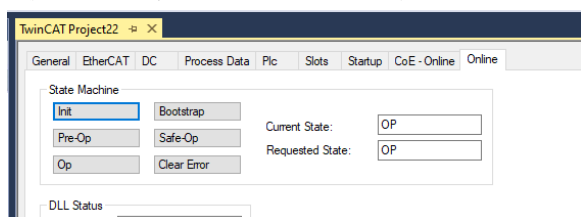
- (3) 已添加“Module 2(csv-axis)”



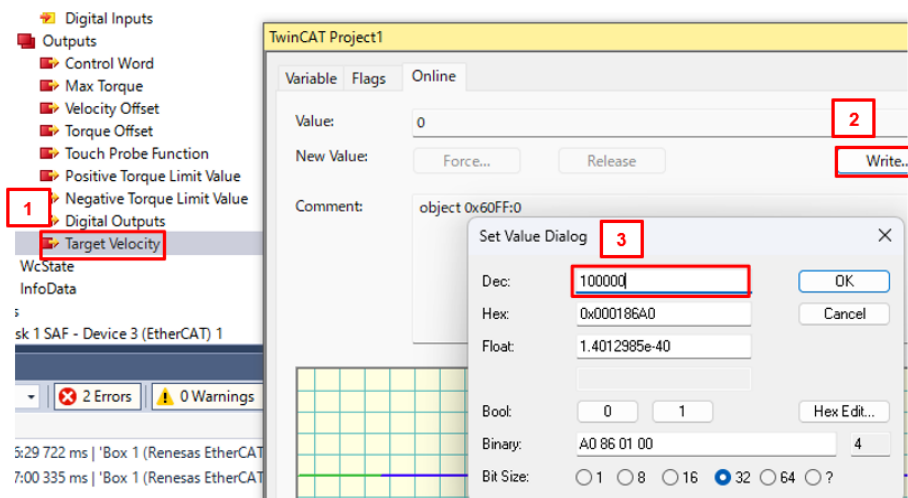
- (4) 点击 “Reload Devices”。



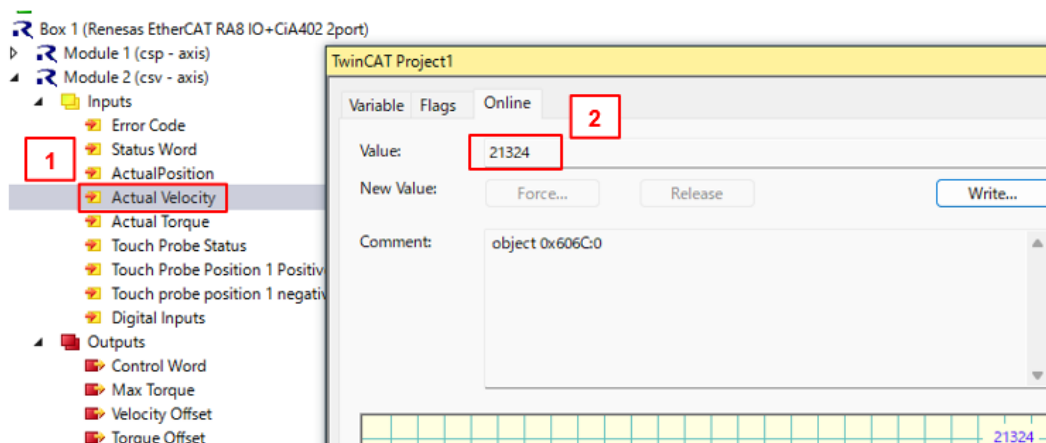
- (5) 执行5.3.2步骤，切换CiA402状态，确认进入“Operation Enabled”。



(6) 设置“Target Velocity”值为100000。



(7) "Actual Velocity" 会变化到100000。



6. CiA402 协议

6.1 操作模式

支持以下操作模式

Operation Mode	Support
Profile position mode	No
Velocity mode (frequency converter)	No
Profile velocity mode	No
Profile torque mode	No
Homing mode	No
Interpolated position mode	No
Cyclic synchronous position mode	Yes
Cyclic synchronous velocity mode	Yes
Cyclic synchronous torque mode	No
Cyclic synchronous torque mode with commutation angle	No
Manufacturer specific mode	No

6.2 状态切换

支持以下状态机切换

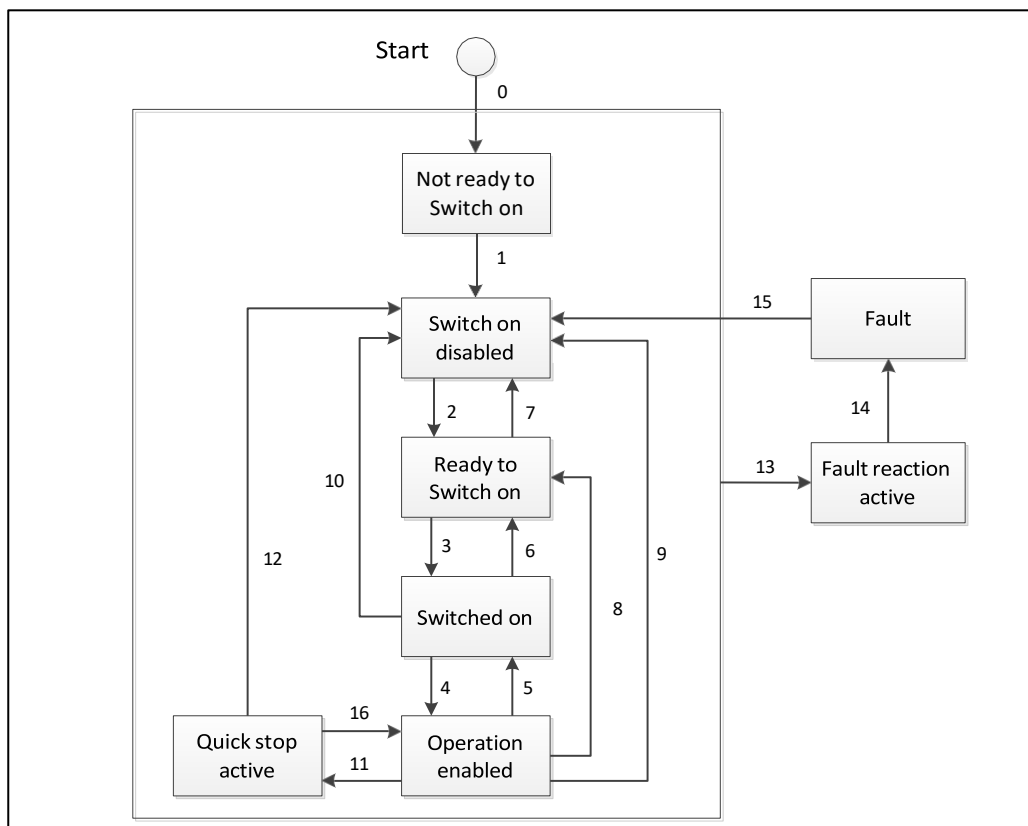


Figure 7.2 CiA402 State Transition Diagram

6.3 对象字典

支持以下对象字典

Operation Mode	Object Name	INDEX	Category	Access	Data Type	PDO Mapping
Cyclic synchronous position mode + Cyclic synchronous velocity mode	Position actual value	0x6064	Mandatory	ro	INT32	Yes
	Following error window	0x6065	Optional	rw	UINT32	No
	Velocity actual value	0x606C	Conditional	ro	INT32	Yes
	Max torque	0x6072	Optional	rw	UINT16	Yes
	Torque actual value	0x6077	Conditional	ro	INT16	Yes
	Target position	0x607A	Optional	rw	INT32	Yes
	Software position limit	0x607D	Optional	c,rw	INT32	No
	Position offset	0x60B0	Optional	rw	INT32	Yes
	Velocity offset	0x60B1	Optional	rw	INT32	Yes
	Torque offset	0x60B2	Optional	rw	INT16	Yes
	Following error actual value	0x60F4	Optional	ro	INT32	Yes
	Target velocity	0x60FF	Conditional	rw	INT32	Yes

Function Group	Object Name	INDEX	Category	Access	Data Type	PDO Mapping
Torque Limiting	Positive torque limit value	0x60E0	Conditional	rw	UINT16	Yes
	Negative torque limit value	0x60E1	Conditional	rw	UINT16	Yes
Homing	Home Offset	0x607C	Optional	rw	INT32	No
	Homing speeds	0x6099	Conditional	c,rw	UINT32	No
Touch Probe	Touch probe function	0x60B8	Optional	rw	UINT16	Yes
	Touch probe status	0x60B9	Optional	ro	UINT16	Yes
	Touch probe pos 1 pos value	0x60BA	Optional	ro	INT32	Yes
	Touch probe pos 2 pos value	0x60BC	Optional	ro	INT32	Yes
Gear ratio	Gear ratio	0x6091	Optional	c,rw	UINT32	No
Other object	Object Name	INDEX	Category	Access	Data Type	PDO Mapping
Controlling the power drive system	Error code	0x603F	Optional	ro	UINT16	Yes
	Controlword	0x6040	Mandatory	rw	UINT16	Yes
	Statusword	0x6041	Mandatory	ro	UINT16	Yes
	Quick stop option code	0x605A	Optional	rw	INT16	No
	Shutdown option code	0x605B	Optional	rw	INT16	No
	Disable operation option code	0x605C	Optional	rw	INT16	No
	Halt option code	0x605D	Optional	rw	INT16	No
	Fault reaction option code	0x605E	Optional	rw	INT16	No
	Modes of operation	0x6060	Optional	rw	INT8	Yes
	Modes of operation disp	0x6061	Optional	ro	INT8	Yes
	Supported drive modes	0x6502	Mandatory	ro	INT32	No
General object	Motor type	0x6402	Optional	rw	INT16	No
Position control function	Position demand value	0x6062	Optional	ro	INT32	No
	Position actual internal value	0x6063	Optional	ro	INT32	No
	Position window	0x6067	Optional	rw	UINT32	No
Optional application FE	Digital inputs	0x60FD	Optional	ro	UINT32	Yes
	Digital outputs	0x60FE	Optional	c,rw	UINT32	No,Yes

6.4 CiA402相关API

CiA402接口函数

CiA402_StateTransition1	
<u>Description</u> This function is used when state transition 1 has occurred. Describe the operation in the case of the state transition.	
<u>Usage</u> #include "cia402appl.h"	
<u>Parameters</u> TCiA402Axis *pCiA402Axis	
<u>Return Value</u> 0 Normal end 1 Error	
<u>Remark</u>	<u>Remark</u> In the case of error occurrence during processing, exit the function by setting the appropriate values for each object in accordance with the CiA402 standard. If 1 is set to return value, state transition does not occur.
CiA402_StateTransition2	
<u>Description</u> This function is used when state transition 2 has occurred. Describe the operation in the case of the state transition.	
<u>Usage</u> #include "cia402appl.h"	
<u>Parameters</u> TCiA402Axis *pCiA402Axis	
<u>Return Value</u> 0 Normal end 1 Error	
<u>Remark</u>	In the case of error occurrence during processing, exit the function by setting the appropriate values for each object in accordance with the CiA402 standard. If 1 is set to return value, state transition does not occur.

CiA402_StateTransition3	
<u>Description</u>	This function is used when state transition 3 has occurred. Describe the operation in the case of the state transition.
<u>Usage</u>	#include "cia402appl.h"
<u>Parameters</u>	TCiA402Axis *pCiA402Axis
<u>Return Value</u>	0 Normal end 1 Error
<u>Remark</u>	In the case of error occurrence during processing, exit the function by setting the appropriate values for each object in accordance with the CiA402 standard. If 1 is set to return value, state transition does not occur.
CiA402_StateTransition4	
<u>Description</u>	This function is used when state transition 4 has occurred. Describe the operation in the case of the state transition.
<u>Usage</u>	#include "cia402appl.h"
<u>Parameters</u>	TCiA402Axis *pCiA402Axis
<u>Return Value</u>	0 Normal end 1 Error
<u>Remark</u>	In the case of error occurrence during processing, exit the function by setting the appropriate values for each object in accordance with the CiA402 standard. If 1 is set to return value, state transition does not occur.
CiA402_StateTransition5	
<u>Description</u>	This function is used when state transition 5 has occurred. Describe the operation in the case of the state transition.
<u>Usage</u>	#include "cia402appl.h"
<u>Parameters</u>	TCiA402Axis *pCiA402Axis
<u>Return Value</u>	0 Normal end 1 Error
<u>Remark</u>	In the case of error occurrence during processing, exit the function by setting the appropriate values for each object in accordance with the CiA402 standard. If 1 is set to return value, state transition does not occur.

CiA402_StateTransition6	
<u>Description</u>	This function is used when state transition 6 has occurred. Describe the operation in the case of the state transition.
<u>Usage</u>	#include "cia402appl.h"
<u>Parameters</u>	TCiA402Axis *pCiA402Axis
<u>Return Value</u>	0 Normal end 1 Error
<u>Remark</u>	In the case of error occurrence during processing, exit the function by setting the appropriate values for each object in accordance with the CiA402 standard. If 1 is set to return value, state transition does not occur.
CiA402_StateTransition7	
<u>Description</u>	This function is used when state transition 7 has occurred. Describe the operation in the case of the state transition.
<u>Usage</u>	#include "cia402appl.h"
<u>Parameters</u>	TCiA402Axis *pCiA402Axis
<u>Return Value</u>	0 Normal end 1 Error
<u>Remark</u>	In the case of error occurrence during processing, exit the function by setting the appropriate values for each object in accordance with the CiA402 standard. If 1 is set to return value, state transition does not occur.
CiA402_StateTransition8	
<u>Description</u>	This function is used when state transition 8 has occurred. Describe the operation in the case of the state transition.
<u>Usage</u>	#include "cia402appl.h"
<u>Parameters</u>	TCiA402Axis *pCiA402Axis
<u>Return Value</u>	0 Normal end 1 Error
<u>Remark</u>	In the case of error occurrence during processing, exit the function by setting the appropriate values for each object in accordance with the CiA402 standard. If 1 is set to return value, state transition does not occur.

CiA402_StateTransition9	
<u>Description</u>	This function is used when state transition 9 has occurred. Describe the operation in the case of the state transition.
<u>Usage</u>	#include "cia402appl.h"
<u>Parameters</u>	TCiA402Axis *pCiA402Axis
<u>Return Value</u>	0 Normal end 1 Error
<u>Remark</u>	In the case of error occurrence during processing, exit the function by setting the appropriate values for each object in accordance with the CiA402 standard. If 1 is set to return value, state transition does not occur.
CiA402_StateTransition10	
<u>Description</u>	This function is used when state transition 10 has occurred. Describe the operation in the case of the state transition.
<u>Usage</u>	#include "cia402appl.h"
<u>Parameters</u>	TCiA402Axis *pCiA402Axis
<u>Return Value</u>	0 Normal end 1 Error
<u>Remark</u>	In the case of error occurrence during processing, exit the function by setting the appropriate values for each object in accordance with the CiA402 standard. If 1 is set to return value, state transition does not occur.
CiA402_StateTransition11	
<u>Description</u>	This function is used when state transition 11 has occurred. Describe the operation in the case of the state transition.
<u>Usage</u>	#include "cia402appl.h"
<u>Parameters</u>	TCiA402Axis *pCiA402Axis
<u>Return Value</u>	0 Normal end 1 Error
<u>Remark</u>	In the case of error occurrence during processing, exit the function by setting the appropriate values for each object in accordance with the CiA402 standard. If 1 is set to return value, state transition does not occur.

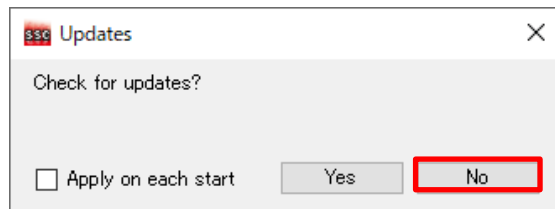
CiA402_StateTransition12	
<u>Description</u>	This function is used when state transition 12 has occurred. Describe the operation in the case of the state transition.
<u>Usage</u>	#include "cia402appl.h"
<u>Parameters</u>	TCiA402Axis *pCiA402Axis
<u>Return Value</u>	0 Normal end 1 Error
<u>Remark</u>	In the case of error occurrence during processing, exit the function by setting the appropriate values for each object in accordance with the CiA402 standard. If 1 is set to return value, state transition does not occur.
CiA402_LocalError	
<u>Description</u>	This function is used when state transition 13 has occurred. Describe the operation in the case of the state transition.
<u>Usage</u>	#include "cia402appl.h"
<u>Parameters</u>	UINT16 ErrorCode
<u>Return Value</u>	None
<u>Remark</u>	If the error corresponding to state transition 13 occurs, call this function after processing required and saving data at error location.
CiA402_StateTransition14	
<u>Description</u>	This function is used when state transition 14 has occurred. Describe the operation in the case of the state transition.
<u>Usage</u>	#include "cia402appl.h"
<u>Parameters</u>	TCiA402Axis *pCiA402Axis
<u>Return Value</u>	0 Normal end 1 Error
<u>Remark</u>	In the case of error occurrence during processing, exit the function by setting the appropriate values for each object in accordance with the CiA402 standard. If 1 is set to return value, state transition does not occur.

CiA402_StateTransition15	
	<u>Description</u> This function is used when state transition 15 has occurred. Describe the operation in the case of the state transition.
	<u>Usage</u> #include "cia402appl.h"
	<u>Parameters</u> TCiA402Axis *pCiA402Axis
	<u>Return Value</u> 0 Normal end 1 Error
	<u>Remark</u> In the case of error occurrence during processing, exit the function by setting the appropriate values for each object in accordance with the CiA402 standard. If 1 is set to return value, state transition does not occur.
CiA402_StateTransition16	
	<u>Description</u> This function is used when state transition 16 has occurred. Describe the operation in the case of the state transition.
	<u>Usage</u> #include "cia402appl.h"
	<u>Parameters</u> TCiA402Axis *pCiA402Axis
	<u>Return Value</u> 0 Normal end 1 Error
	<u>Remark</u> In the case of error occurrence during processing, exit the function by setting the appropriate values for each object in accordance with the CiA402 standard. If 1 is set to return value, state transition does not occur.
APPL_MOTOR_MotionControl_Main	
	<u>Description</u> Implement the motion control code when the state of CiA402 FSA is "Operation enabled". Describe the process for each mode of operation.
	<u>Usage</u> #include "cia402appl.h"
	<u>Parameters</u> TCiA402Axis *pCiA402Axis
	<u>Return Value</u> 0 Normal end 1 Error
	<u>Remark</u> At the initial state, this function is described in "main.c" and calls "CiA402_DummyMotionControl" function for reference.

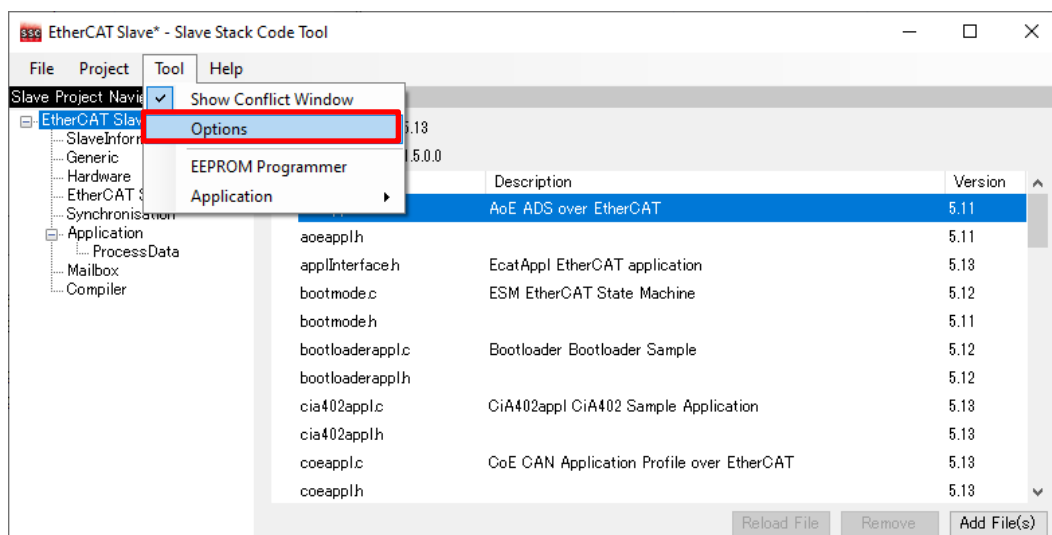
7. 附录

7.1 附录 A:使用SSC工具注意事项

1. 使用管理员权限打开SSC软件
2. 打开SSC软件时，该提示选择No



3. 选项设置



4. 不要勾选

Slave Stack Code Tool | Options

Generic Editor Create Files Configurations

On startup

☐ Open last project

☐ Check for updates

EtherCAT vendor information

Vendor name Renesas Electronics Corporation

Vendor ID 0x766

Default Patch

☐ Apply on new Projects

Open

5. 不要勾选

Slave Stack Code Tool | Options

Generic Editor Create Files Configurations

☐ Add comments if obsolete code was skipped

☐ Create device description (ESI)

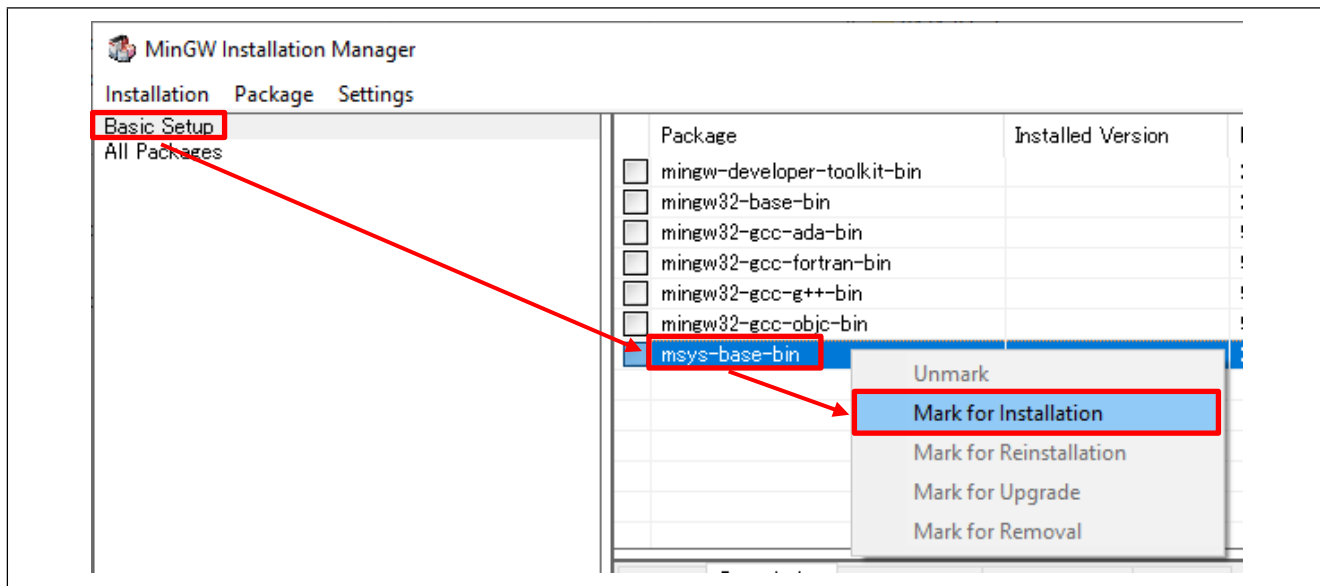
☐ Create documentation

7.2 附录 B: 怎么安装patch工具

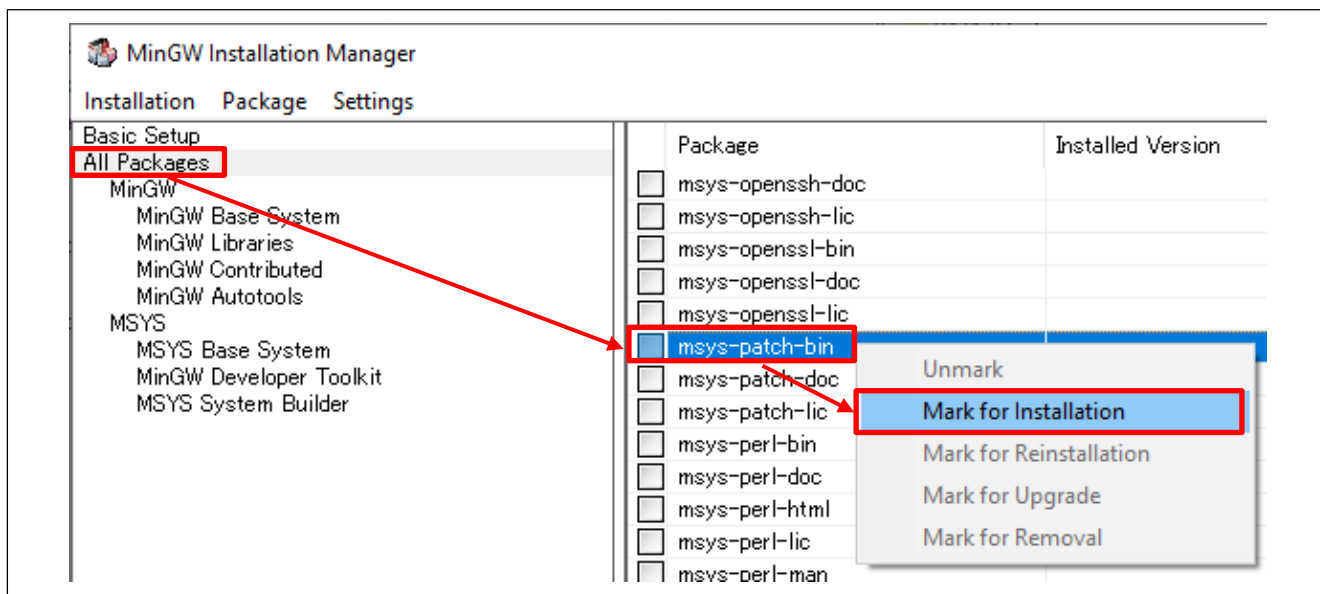
1. 下载“mingw-get-setup.exe”。

<https://sourceforge.net/projects/mingw/postdownload>

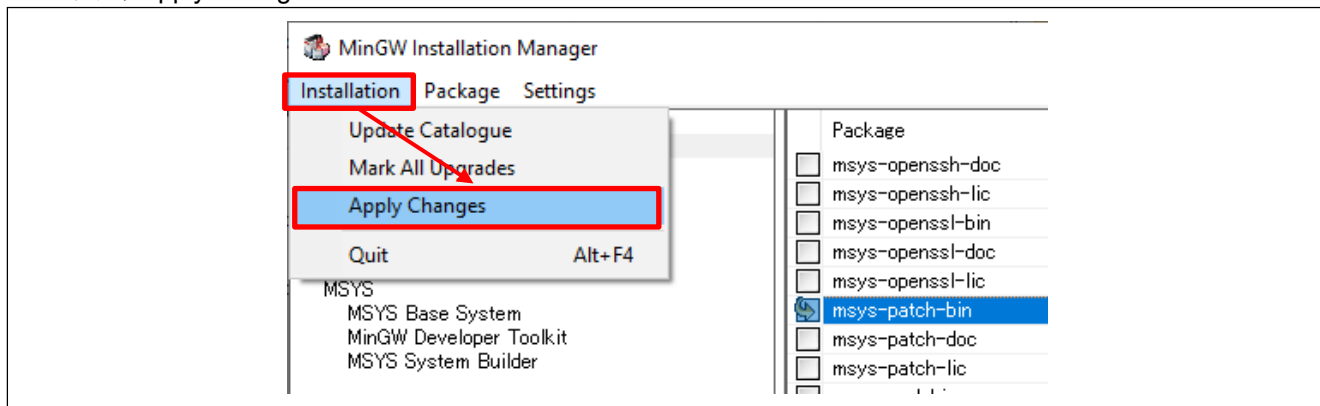
2. 运行“mingw-get-setup.exe”，安装“MinGw-installation-manager”
3. 选择“Basic Setup”，右击“msys-base-bin”然后选择“Mark for Installation”。



4. 选择All Packages，右击msys-patch-bin，选择Mark for Installation。



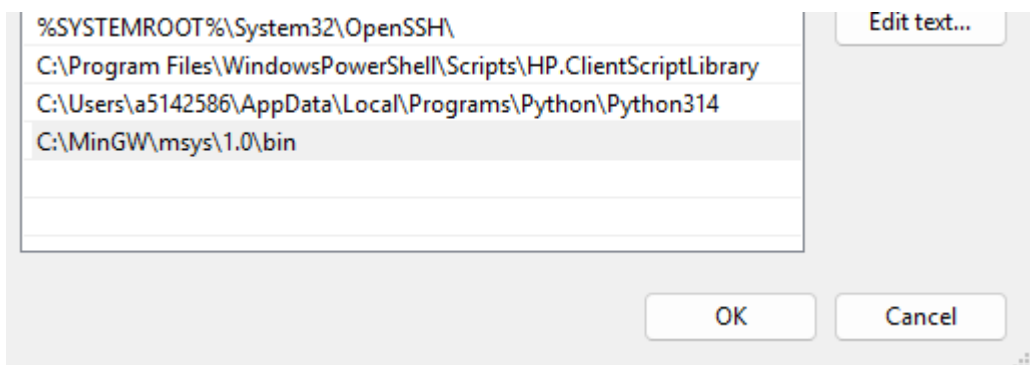
5. 点击“Apply Changes”



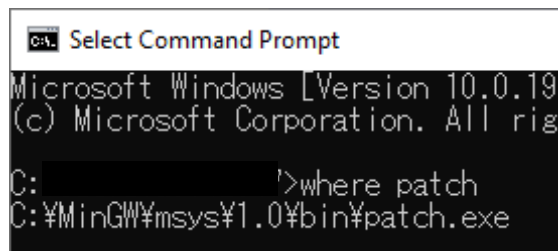
6. 打开“Schedule of Pending Actions”窗口，点击“Apply”

7. 出现“All changes were applied successfully; you may now close this dialog.” 表示patch. 安装成功

8. 添加环境变量，重启电脑



9. 查询是否安装成功

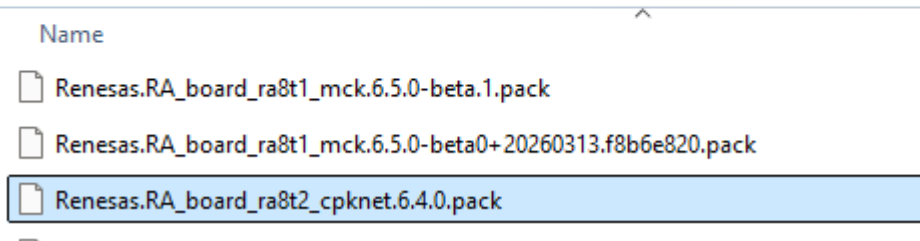


7.3 附录 C:重新创建工程的FSP配置步骤（不导入本样例程序）

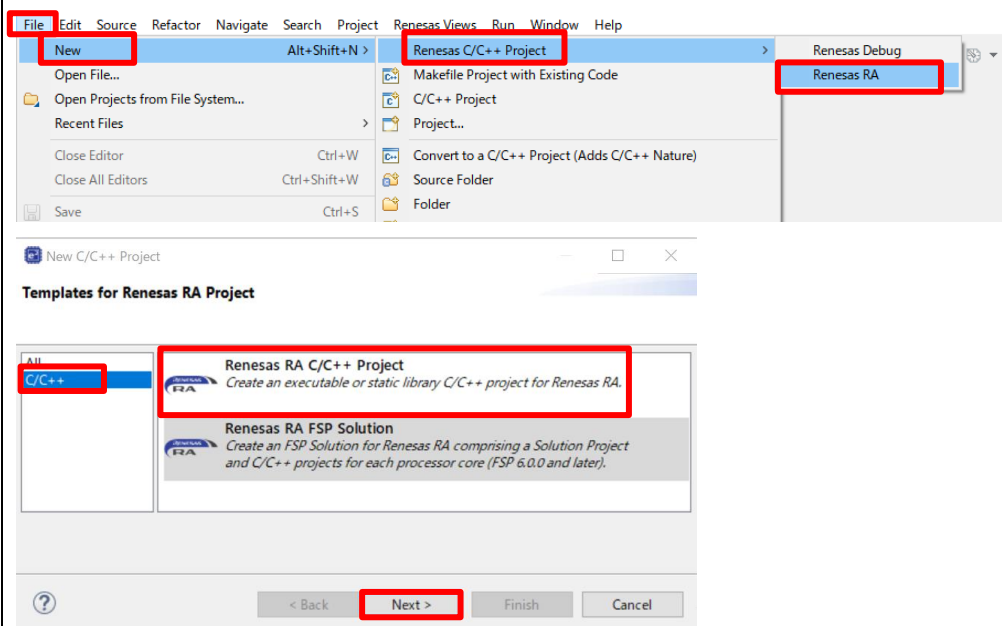
1.0

安装 CPK RA8T2 BSP 文件，将 Renesas.RA_board_ra8t2_cpknet.6.4.0.pack 文件拷贝到 e2 studio 的安装目录中。

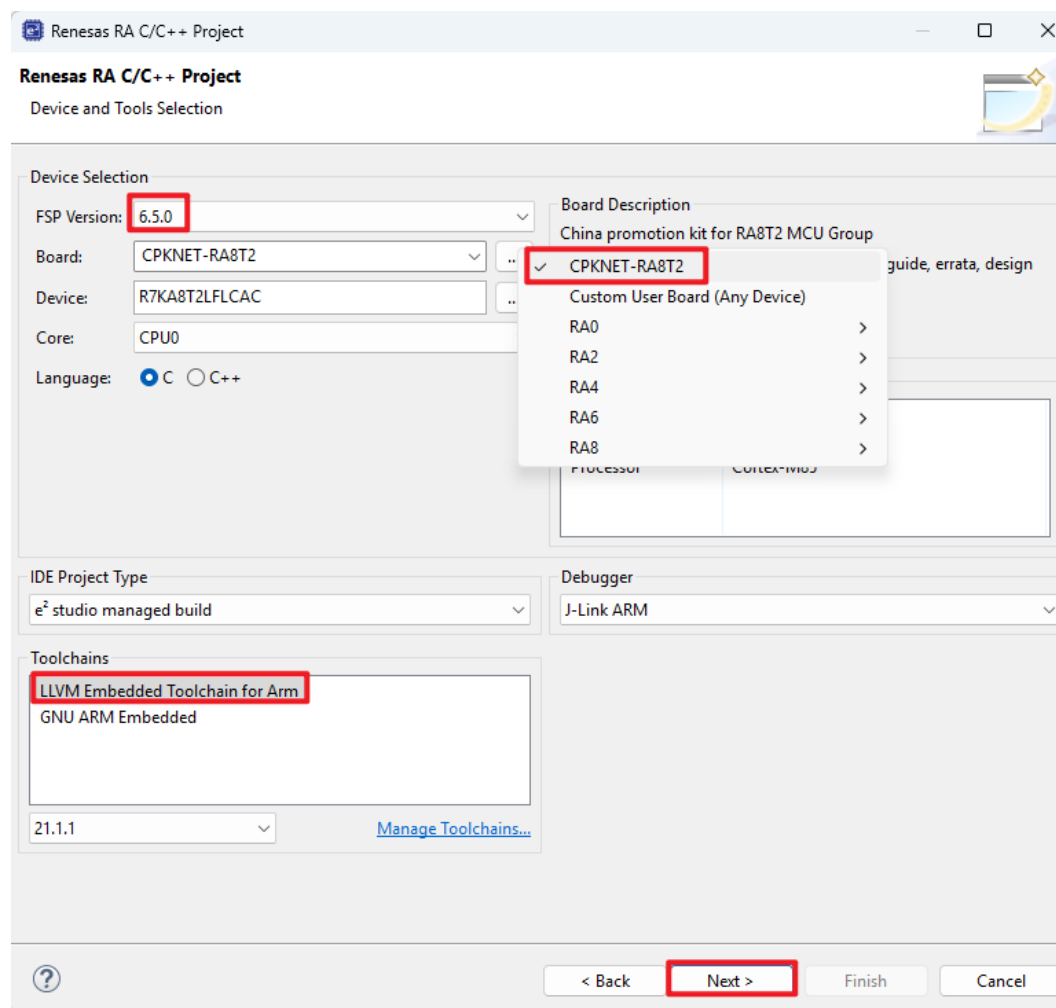
```
> e2_studio > internal > projectgen > ra > packs
```



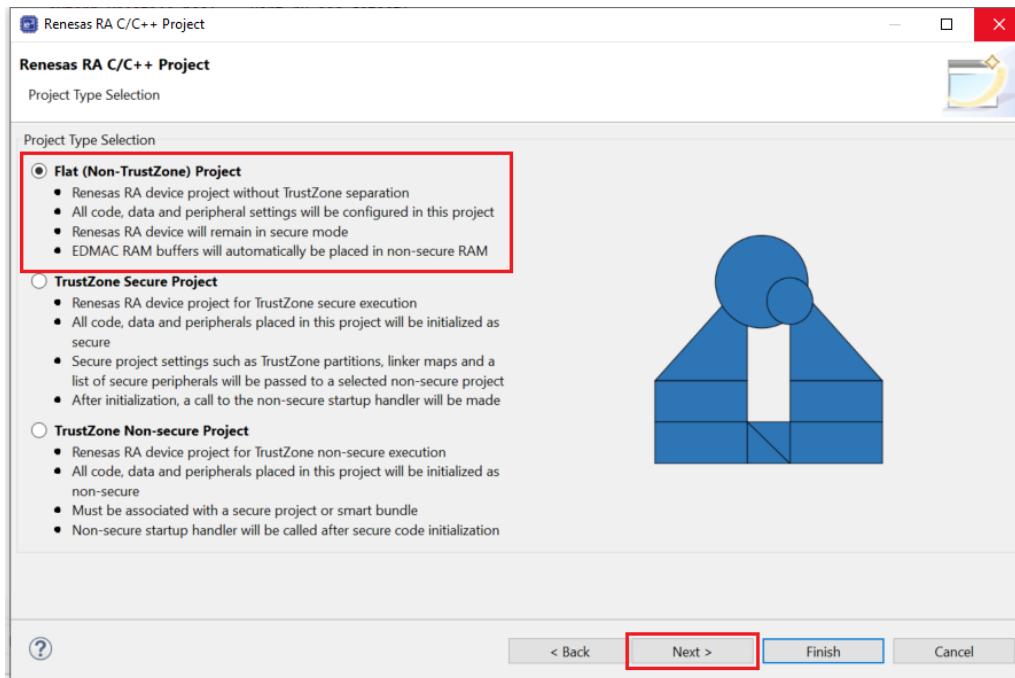
首先创建 CPU0 工程，运行 e² studio 新建 C/C++ Project. 点击 **File > New > Renesas C/C++ Project > Renesas RA** 选择 **Renesas RA C/C++ Project**. 最后点击 **Next>**. 工程命名为 “ecat_io_402_cpknet_ra8t2_cpu0_ep_llvm”，点击 **Next>**.



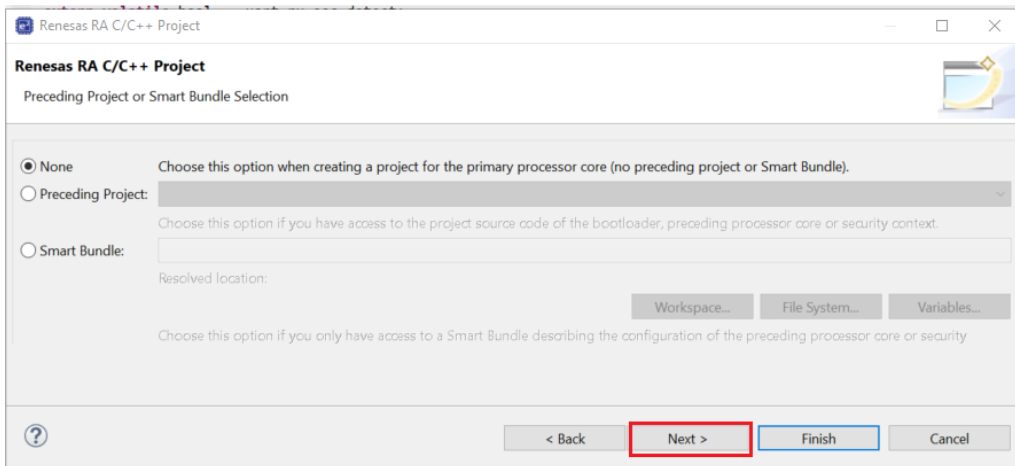
选择 **FSP version 6.5.0,CPU0**, 选择编译器选项设置如下, 然后点击 **Next>**.



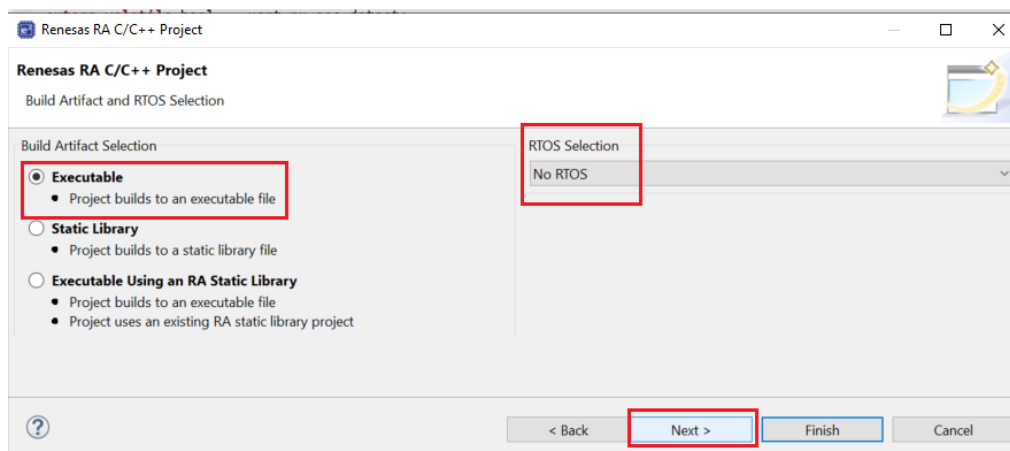
选择 **Flat(Non-TrustZone)Project**，点击 **Next>**



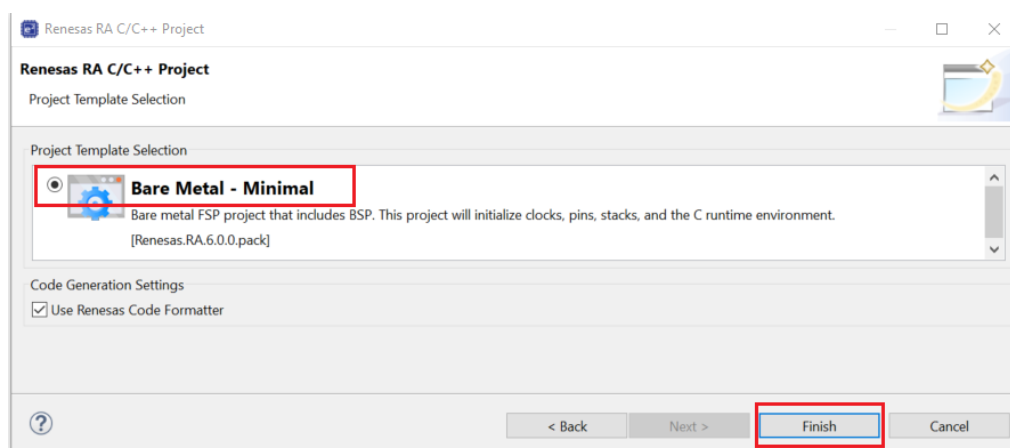
选择 **None**，点击 **Next**



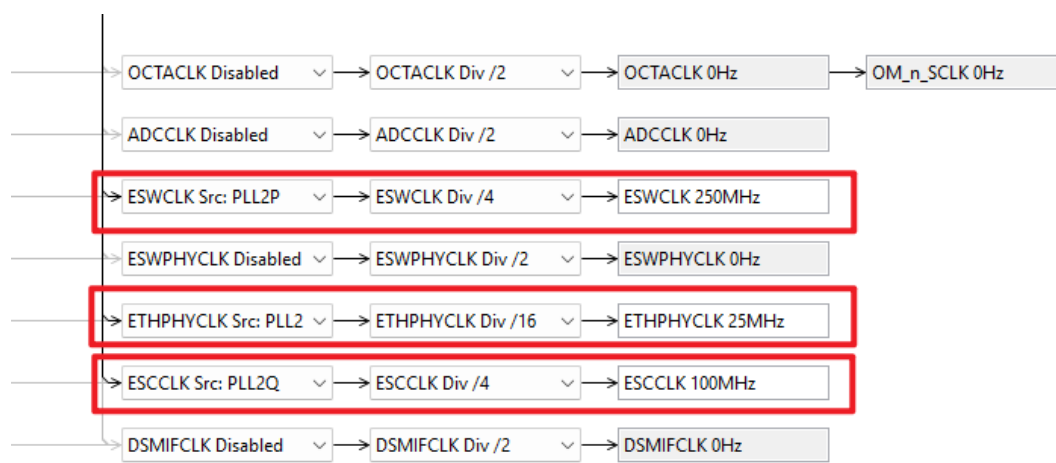
1.6

选择 **Executable**, **No RTOS**, 点击 **Next**

1.7

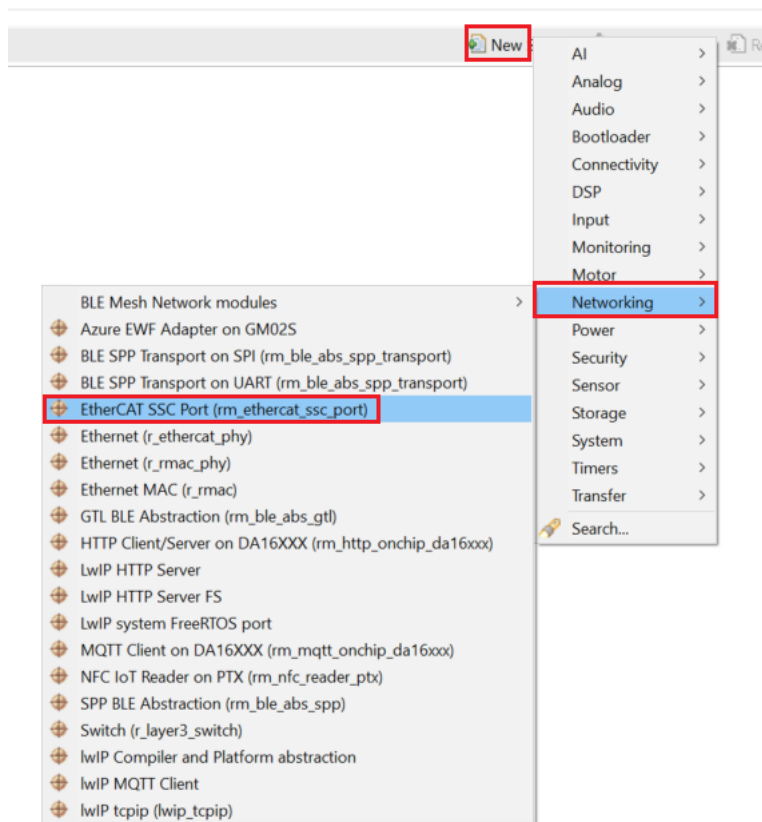
选择 **Bare Metal – Minimal**, 点击 **Finish**

1.8

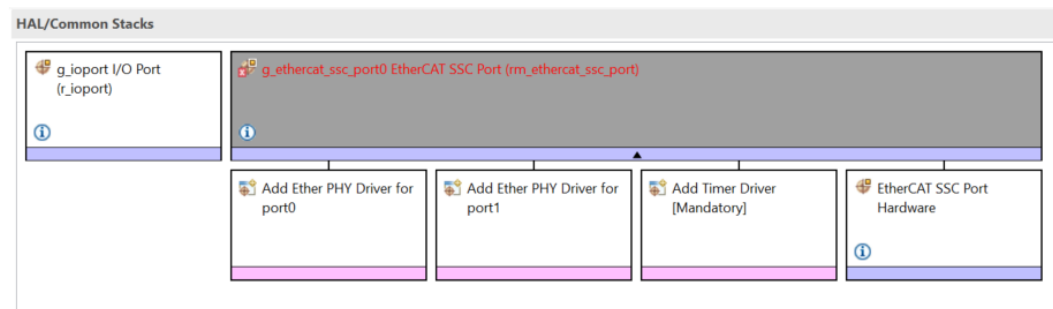
配置时钟, 点击 **Clock**, 修改时钟配置如下:

1.9

添加 ECAT driver，点击 Stack,New->Networking->EtherCAT SSC Port

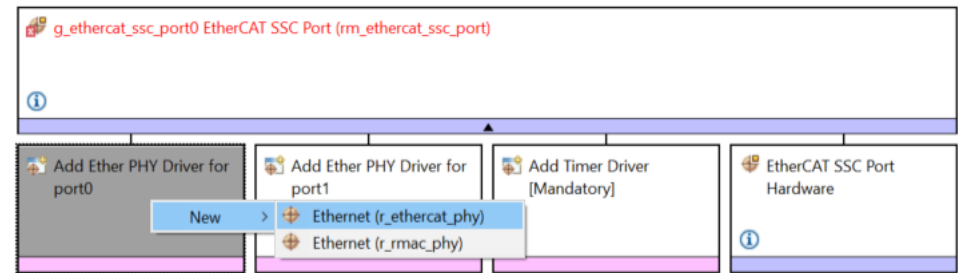


生成配置如下：

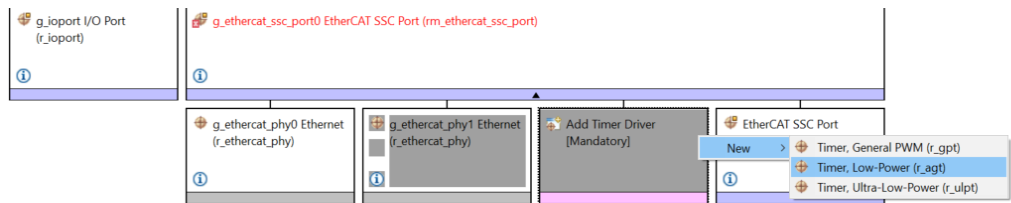


1.10

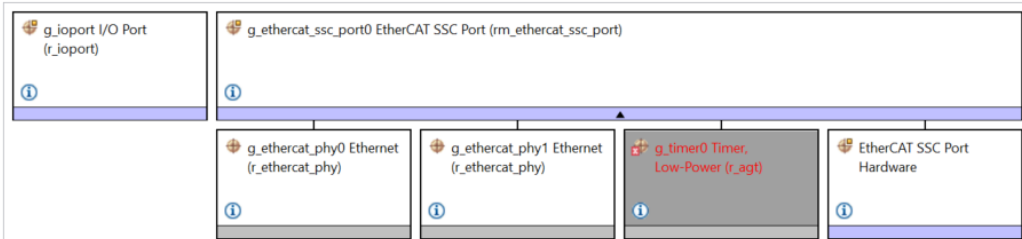
添加 PHY 和 timer，分别点击 Add Ether PHY Driver for port0 和 Add Ether PHY Driver for port1，选择 Ethernet (r_ethercat_phy)



添加 timer，选择 Timer, Low-Power (r_agt)



生成配置如下



1.11

修改各个 Driver 属性，点击 Properties（如配置界面没有 Properties 选项，可通过菜单栏 Window->Show View->other 打开）

点击 g_timer0 Timer，设置中断优先级，此时红色错误消失。

Module g_timer0 Timer, General PWM (r_gpt)	
> General	
> Output	
> Input	
> Pin Polarity	
> Interrupts	
Callback	ethercat_ssc_port_timer_callback
Overflow/Crest Interrupt Priority	Priority 13
Capture/Compare match A Interrupt Priority	Disabled
Capture/Compare match B Interrupt Priority	Disabled
Compare Match C Interrupt Priority	Disabled

1.12

修改 PHY 属性，设置偏移地址为 1

Property	Value
▼ Common	
Parameter Checking	Default (BSP)
Phy Link Polarity0	Active Low
Phy Link Polarity1	Active High
▼ Module g_ethercat_ssc_port0 EtherCAT SSC Port (rm_ethercat_ssc_port)	
▼ PHY	
Reset Signal Hold Time (ms)	1
Reset Wait Time (us)	15000
Offset Address	1
> Delay Time of TX Signal	
> Interrupts	
Name	g_ethercat_ssc_port0
EEPROM Size	Under 32Kbits

点击选择 Add Ether PHY Driver for port0，修改配置如下

▼ Common	
Parameter Checking	Default (BSP)
ICS1894 Target	Disabled
User Own Target	Enabled
▼ Module g_ethercat_phy0 Ethernet (r_ethercat_phy)	
Name	g_ethercat_phy0
Channel	0
PHY-LSI Address	0
PHY-LSI Reset Completion Timeout	0x00020000
Select MII type	MI
PHY-LSI Type	User own PHY
Target Init Function	ethercat_phy_target_yt8512_initialize
Target Link Partner Ability Get Function	ethercat_phy_target_yt8512_is_support_link_partner_ability
MI Register Access Wait-time	8
Flow Control	Disable

接着点击选择 Add Ether PHY Driver for port1，修改配置如下：

▼ Common	
Parameter Checking	Default (BSP)
ICS1894 Target	Disabled
User Own Target	Enabled
▼ Module g_ethercat_phy1 Ethernet (r_ethercat_phy)	
Name	g_ethercat_phy1
Channel	1
PHY-LSI Address	1
PHY-LSI Reset Completion Timeout	0x00020000
Select MII type	MI
PHY-LSI Type	User own PHY
Target Init Function	ethercat_phy_target_yt8512_initialize
Target Link Partner Ability Get Function	ethercat_phy_target_yt8512_is_support_link_partner_ability
MI Register Access Wait-time	8
Flow Control	Disable

1.13

配置引脚，配置 PHY 时钟引脚为 P308。

Pin Selection	Pin Configuration
Type filter text	Name
> CLKOUT:CLKOUT	Pin Group Selection
> CLKOUT:CMPOUT	Value
> ✓ CLKOUT:ETHPHYCLK	Mixed
> ✓ ETHPHYCLK	Operation Mode
> Connectivity:CANFD	Custom
> Connectivity:DSMIF	▼ Input/Output
> Connectivity:ESC	ETHPHYCLK
> Connectivity:ESWM_GMII/MI	Value
> Connectivity:ESWM_RGMII	✓ P308
Pin Function	Lock
Pin Number	Link
Summary	Module name: ETHPHYCLK
BSP	
Clocks	
Pin	
Interrupts	
Event Links	
Linker Sections	
Stacks	
Components	

1.14

配置 ESC 引脚如下:

The screenshot shows the Pin Configuration tool interface. On the left, the 'Pin Selection' pane has a search filter and a tree view. Under 'Connectivity:ESC', 'ESC' is selected and highlighted with a red box. On the right, the 'Pin Configuration' table lists pins and their values. A red box highlights the 'Value' column for pins CAT0_MDC through CAT0_SYNC1.

Name	Value	Lock	Link
Pin Group Selection	Mixed		
Operation Mode	Custom		
Input/Output			
CAT0_MDC	✓ P708		
CAT0_MDIO	✓ P709		
CAT12CLK	✓ P403		
CAT12CDATA	✓ P404		
CATIRQ	✓ PB09		
CATLATCH0	None		
CATLATCH1	None		
CATLEDERR	✓ PB11		
CATLEDRUN	✓ PB08		
CATLEDSTER	✓ PB10		
CATLINKACT0	✓ PB12		
CATLINKACT1	✓ PB13		
CATRESETOUT	✓ P711		
CATSYNC0	✓ PB14		
CATSYNC1	✓ P714		

1.15

配置 ESC0 引脚如下:

The screenshot shows the Pin Configuration tool interface. On the left, the 'Pin Selection' pane has a search filter and a tree view. Under 'Connectivity:ESC', 'ESC0' is selected and highlighted with a red box. On the right, the 'Pin Configuration' table lists pins and their values. A red box highlights the 'Value' column for pins CAT0_ERXD0 through CAT0_TX_EN.

Name	Value	Lock	Link
Pin Group Selection	Mixed		
Operation Mode	Custom		
Input/Output			
CAT0_ERXD0	✓ P702		
CAT0_ERXD1	✓ P701		
CAT0_ERXD2	✓ P700		
CAT0_ERXD3	✓ P406		
CAT0_ETXD0	✓ PB00		
CAT0_ETXD1	✓ PB02		
CAT0_ETXD2	✓ PB03		
CAT0_ETXD3	✓ PB04		
CAT0_LINKSTA	✓ P402		
CAT0_RX_CLK	✓ P703		
CAT0_RX_DV	✓ P405		
CAT0_RX_ER	✓ P704		
CAT0_TX_CLK	✓ PB01		
CAT0_TX_EN	✓ P705		

1.16

配置 ESC1 引脚如下:

The screenshot shows the Pin Configuration tool interface. On the left, the 'Pin Selection' pane has a search filter and a tree view. Under 'Connectivity:ESC', 'ESC1' is selected and highlighted with a red box. On the right, the 'Pin Configuration' table lists pins and their values. A red box highlights the 'Value' column for pins CAT1_ERXD0 through CAT1_TX_EN.

Name	Value	Lock	Link
Pin Group Selection	Mixed		
Operation Mode	Custom		
Input/Output			
CAT1_ERXD0	✓ P906		
CAT1_ERXD1	✓ P907		
CAT1_ERXD2	✓ P908		
CAT1_ERXD3	✓ P909		
CAT1_ETXD0	✓ P307		
CAT1_ETXD1	✓ P306		
CAT1_ETXD2	✓ P305		
CAT1_ETXD3	✓ P304		
CAT1_LINKSTA	✓ P511		
CAT1_RX_CLK	✓ P905		
CAT1_RX_DV	✓ P206		
CAT1_RX_ER	✓ P312		
CAT1_TX_CLK	✓ P311		
CAT1_TX_EN	✓ P310		

1.17	修改堆栈大小: <table><tr><td>> R7KA8T2LFLCAC</td><td></td></tr><tr><td>> RA8T2</td><td></td></tr><tr><td>> RA8T2 Device Options</td><td></td></tr><tr><td>> RA8T2 Family</td><td></td></tr><tr><td>▼ RA Common</td><td></td></tr><tr><td> Main stack size (bytes)</td><td>0x4000</td></tr><tr><td> Heap size (bytes)</td><td>0x8000</td></tr><tr><td> Bootloader Secondary XIP</td><td>Disabled</td></tr><tr><td> MCU Vcc (mV)</td><td>3300</td></tr><tr><td> Parameter checking</td><td>Disabled</td></tr><tr><td> Assert Failures</td><td>Return FSP_ERR_ASSERTION</td></tr></table>	> R7KA8T2LFLCAC		> RA8T2		> RA8T2 Device Options		> RA8T2 Family		▼ RA Common		Main stack size (bytes)	0x4000	Heap size (bytes)	0x8000	Bootloader Secondary XIP	Disabled	MCU Vcc (mV)	3300	Parameter checking	Disabled	Assert Failures	Return FSP_ERR_ASSERTION
> R7KA8T2LFLCAC																							
> RA8T2																							
> RA8T2 Device Options																							
> RA8T2 Family																							
▼ RA Common																							
Main stack size (bytes)	0x4000																						
Heap size (bytes)	0x8000																						
Bootloader Secondary XIP	Disabled																						
MCU Vcc (mV)	3300																						
Parameter checking	Disabled																						
Assert Failures	Return FSP_ERR_ASSERTION																						
1.18	至此 FSP 配置完成，保存配置，点击 Generate Project Content																						

